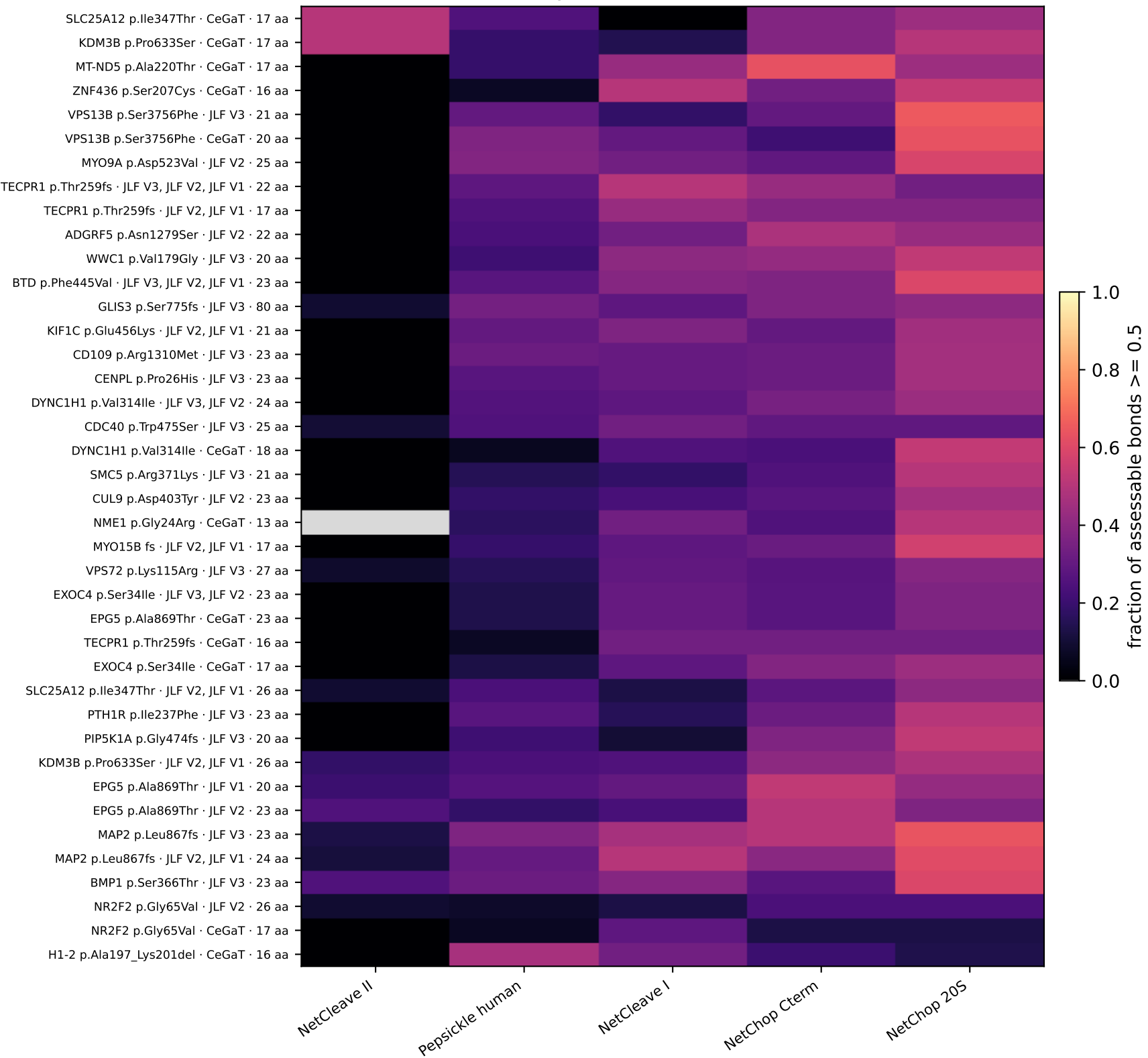
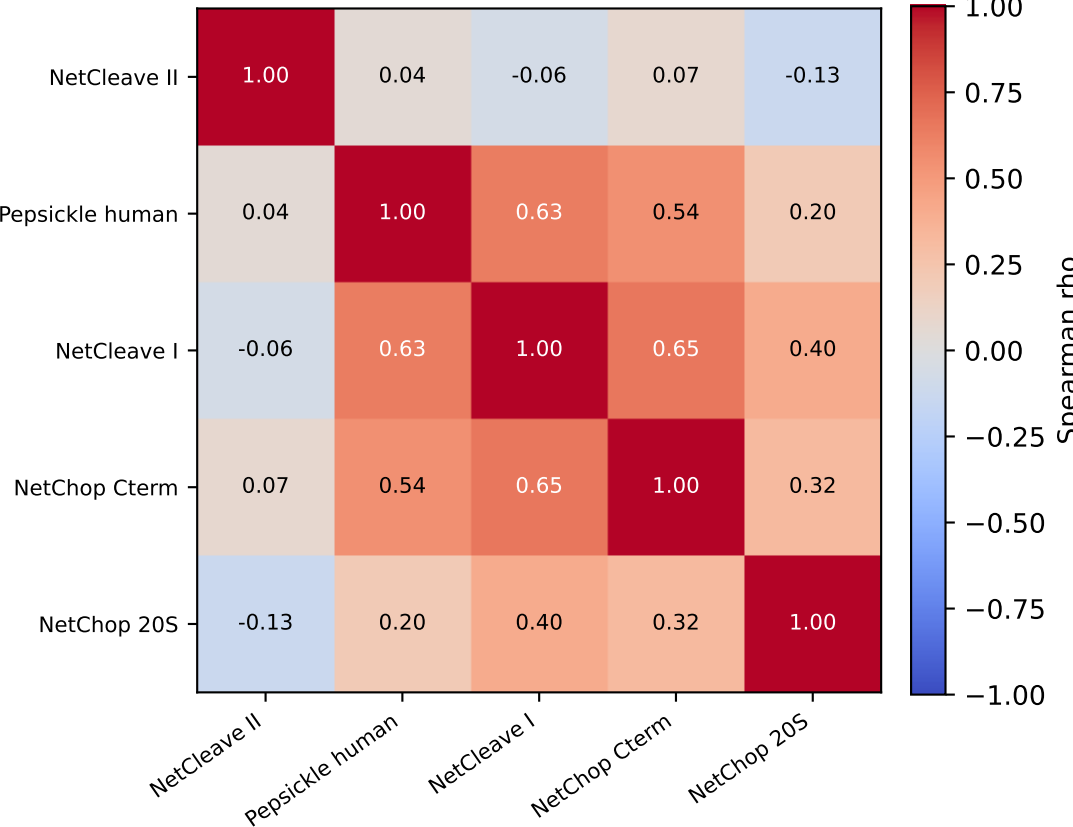


Agreement map for the vaccine SLP cleavage atlas

SLPs clustered by within-model candidate-site fraction



Predictor agreement at shared peptide bonds



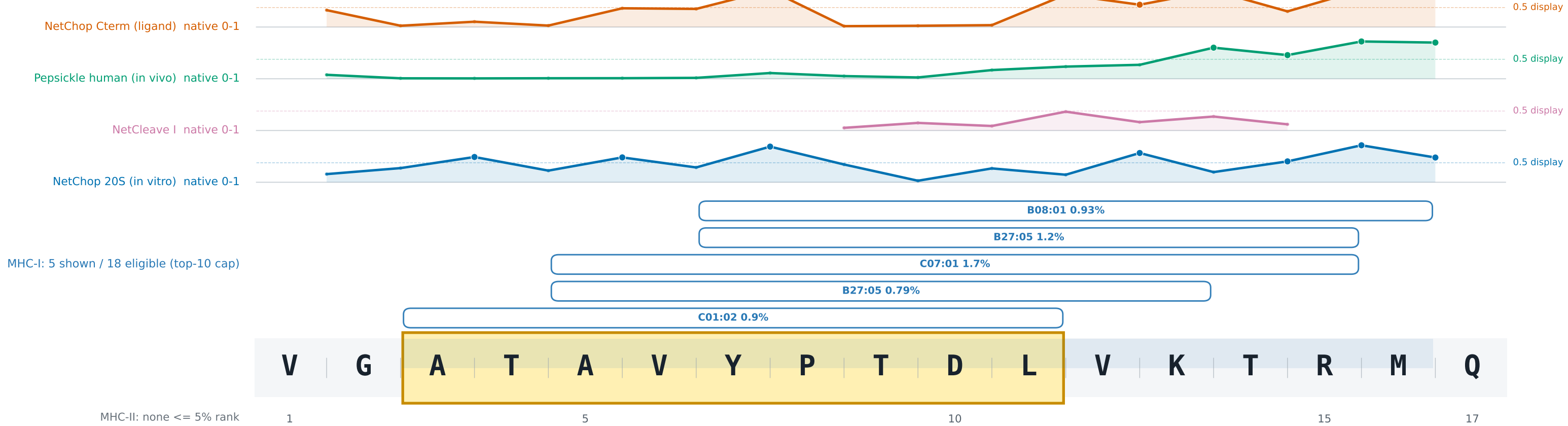
Clustering organizes the atlas; it does not create an ensemble probability. Sequence pages follow the left-panel SLP order. The figure shows human-only Pepsickle, both biologically distinct NetChop modes, NetCleave-I, and NetCleave-II; the near-redundant all-mammal Pepsickle track remains in the tables. The left panel uses a common display threshold on native 0-1 outputs. The right panel correlates native scores only where both models assess the same bond.

SLC25A12 p.Ile347Thr | CeGaT | 17 aa

disclosed minimal epitope ATAVYPTDL at 3-11

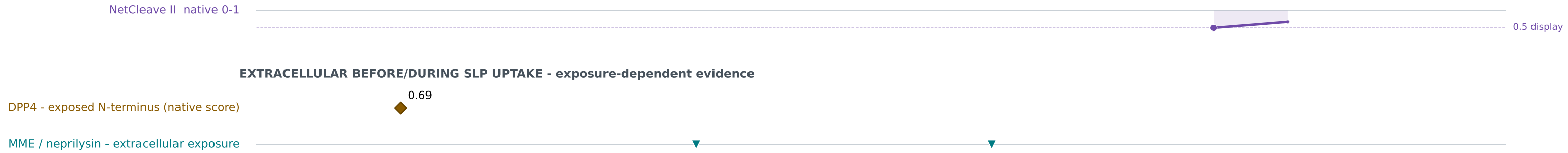
CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

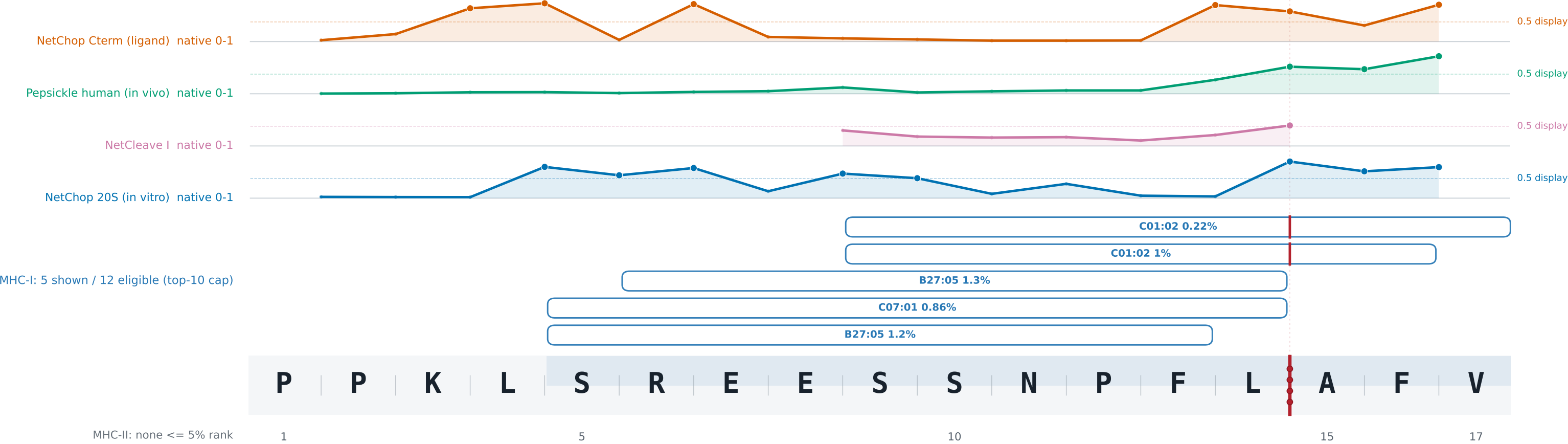
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

KDM3B p.Pro633Ser | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

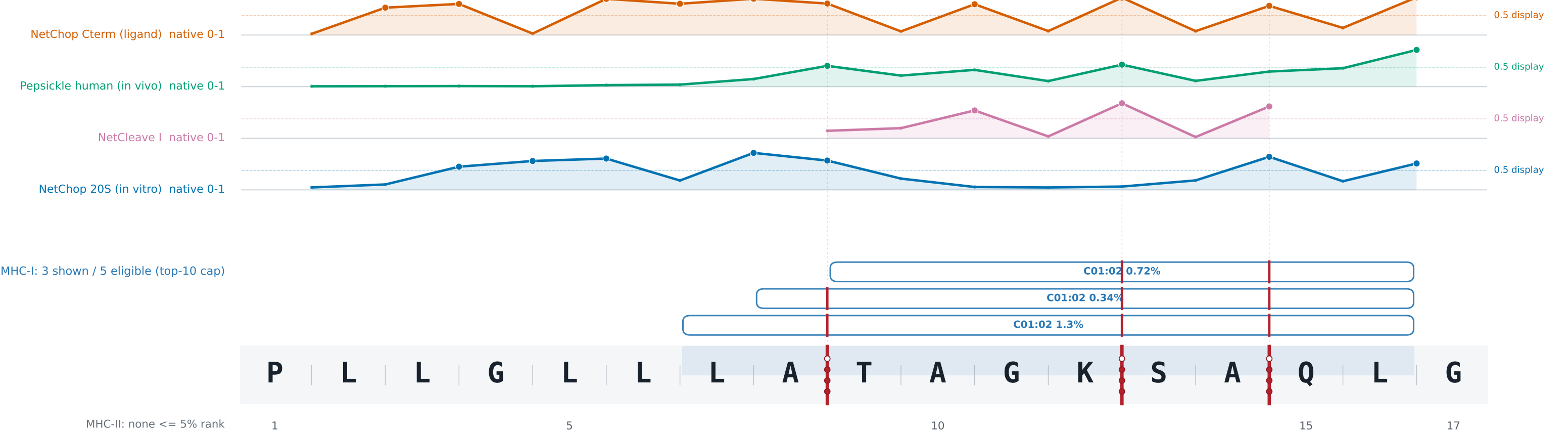
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MT-ND5 p.Ala220Thr | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

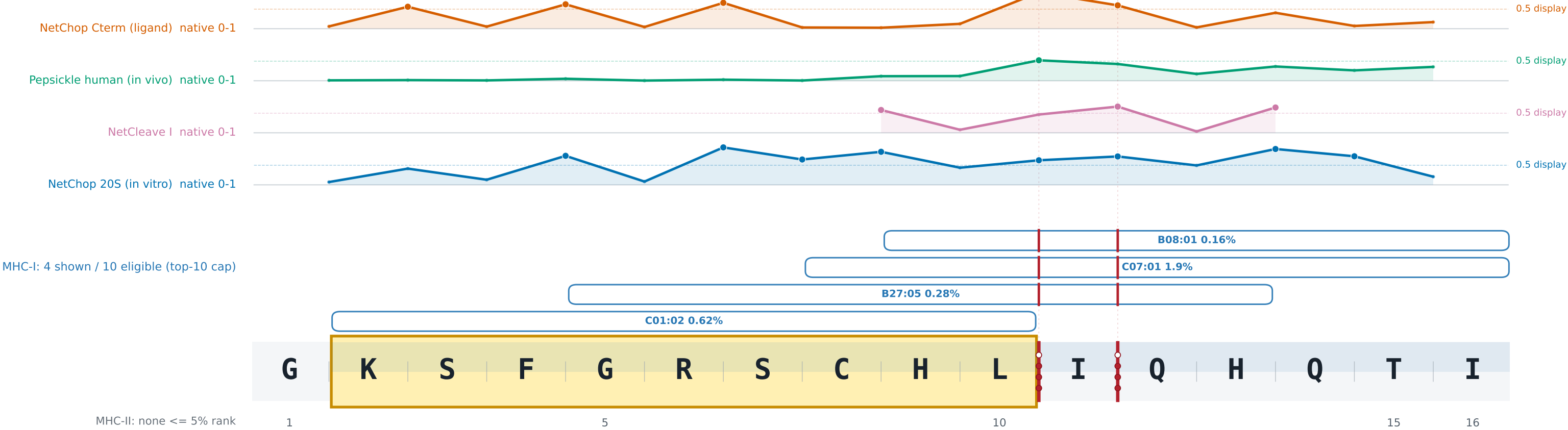
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

ZNF436 p.Ser207Cys | CeGaT | 16 aa

disclosed minimal epitope KSFGRSCHL at 2-10

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

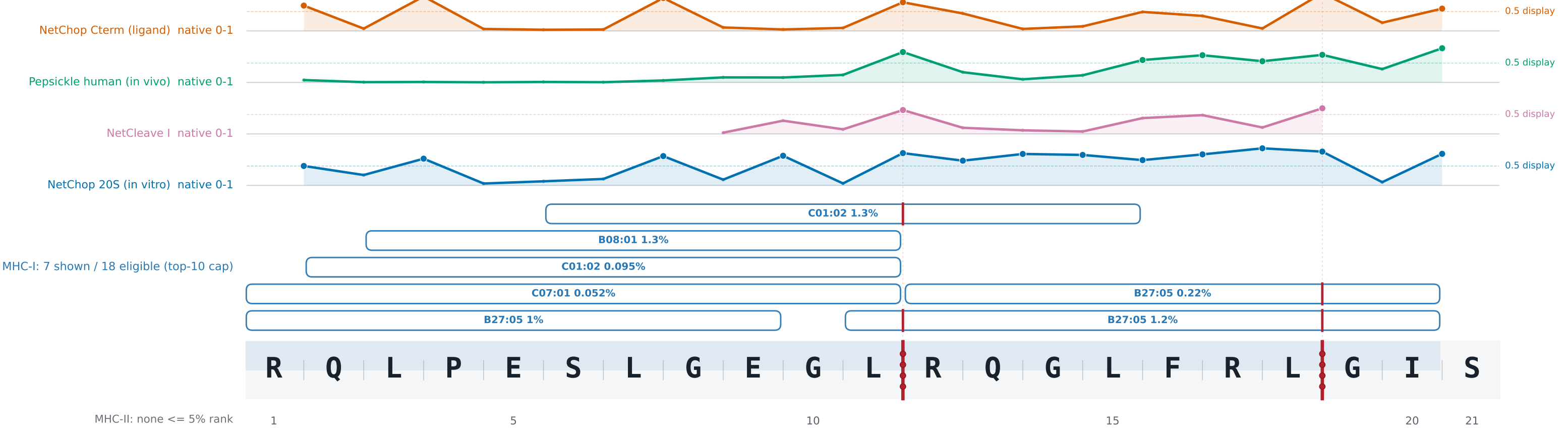
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

VPS13B p.Ser3756Phe | JLF V3 | 21 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

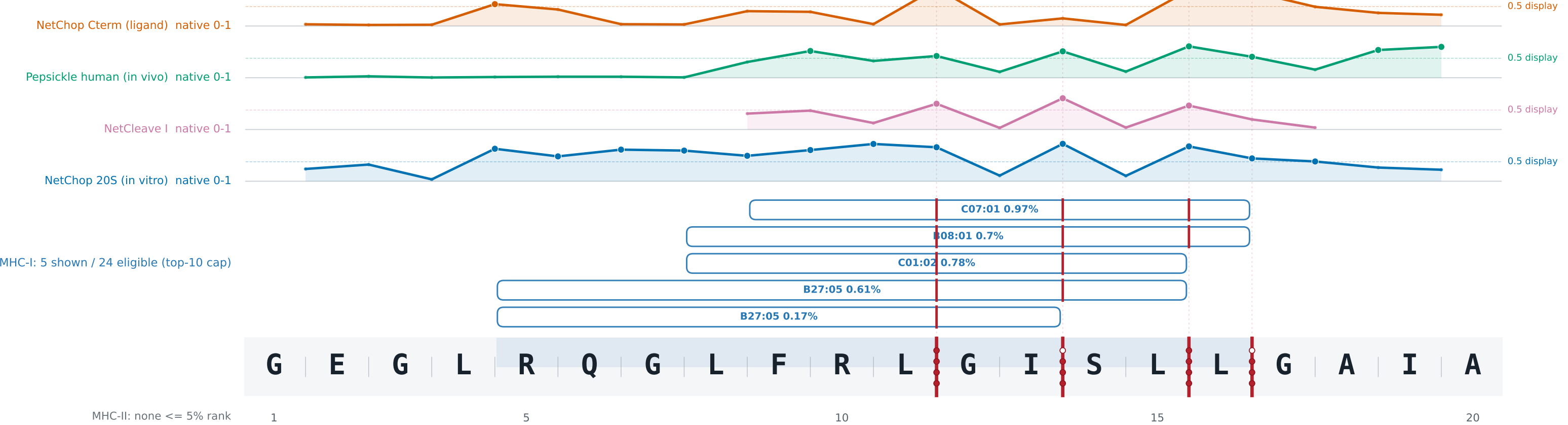
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

VPS13B p.Ser3756Phe | CeGaT | 20 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

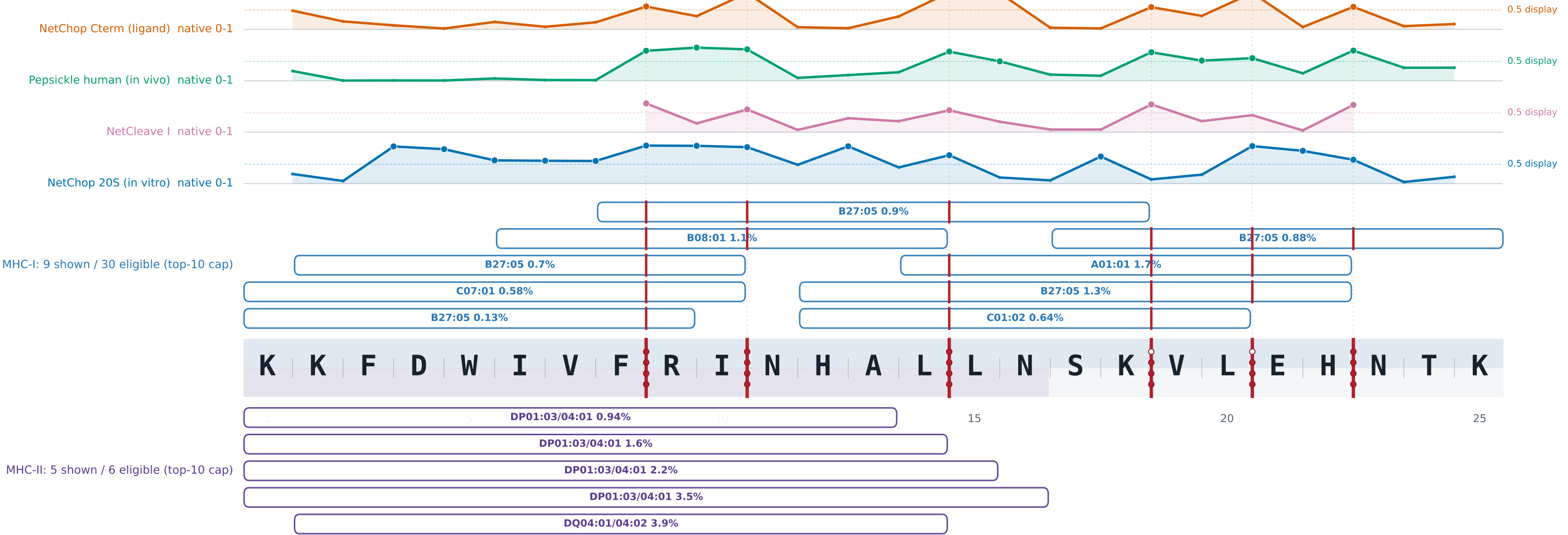
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MYO9A p.Asp523Val | JLF V2 | 25 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

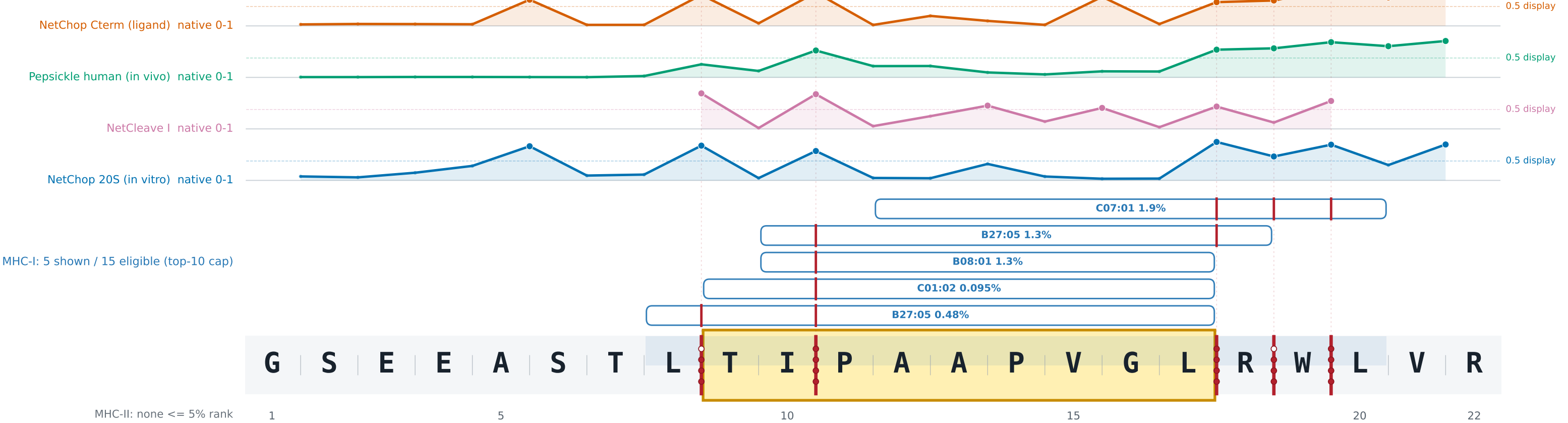
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

TECPR1 p.Thr259fs | JLF V3, JLF V2, JLF V1 | 22 aa

disclosed minimal epitope TIPAAPVGL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

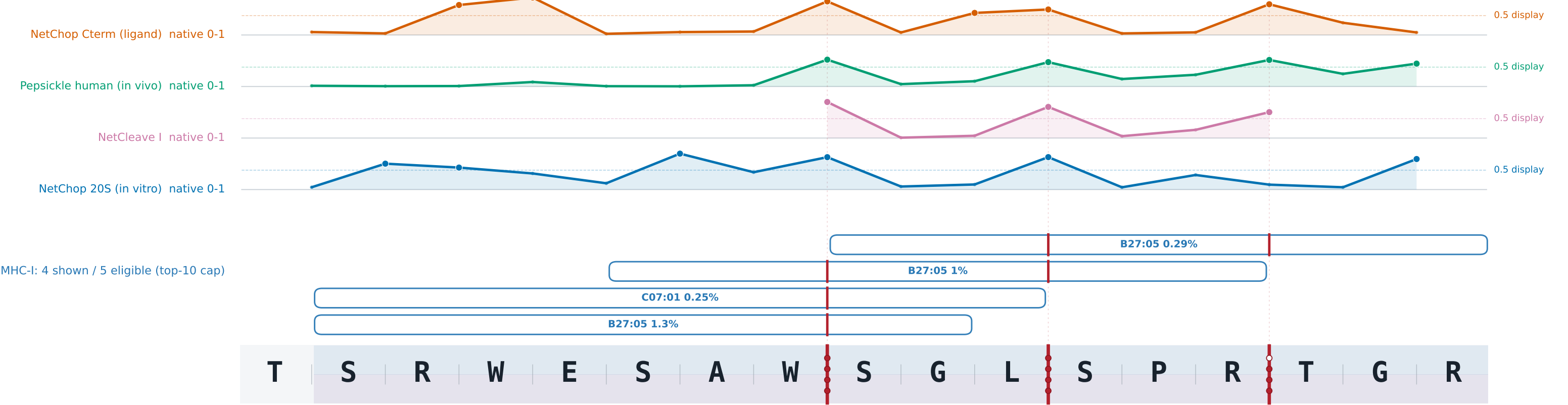


TECPR1 p.Thr259fs | JLF V2, JLF V1 | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

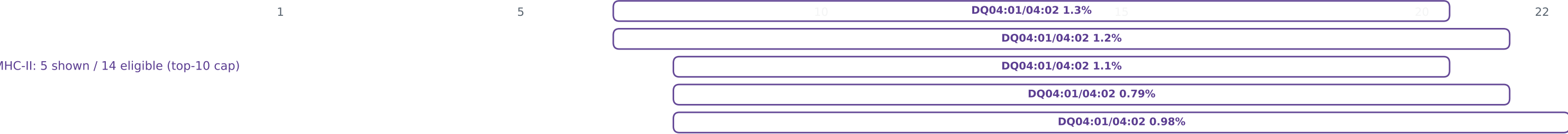
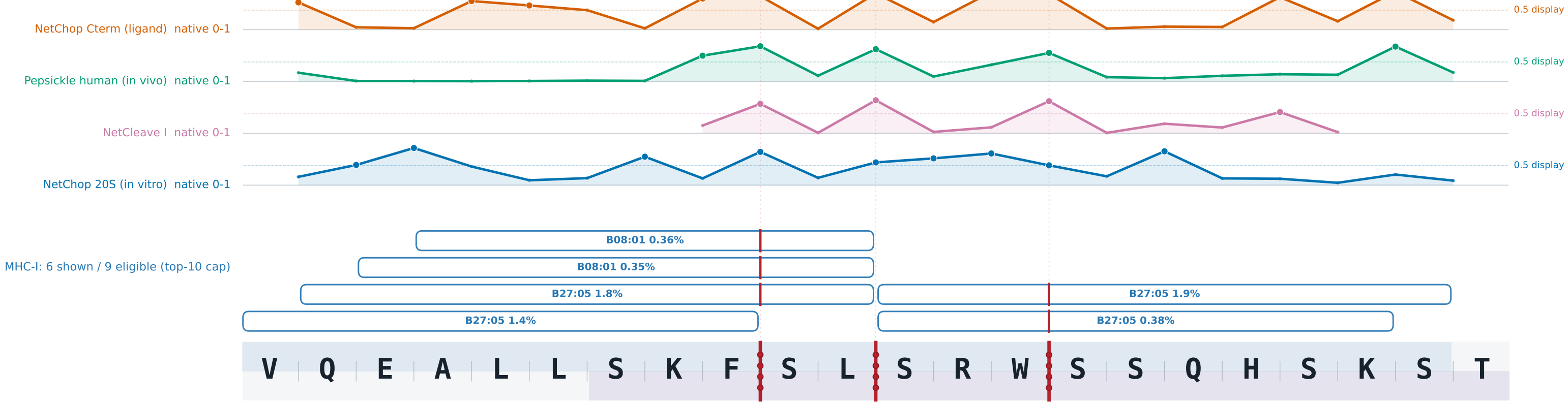
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

ADGRF5 p.Asn1279Ser | JLF V2 | 22 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

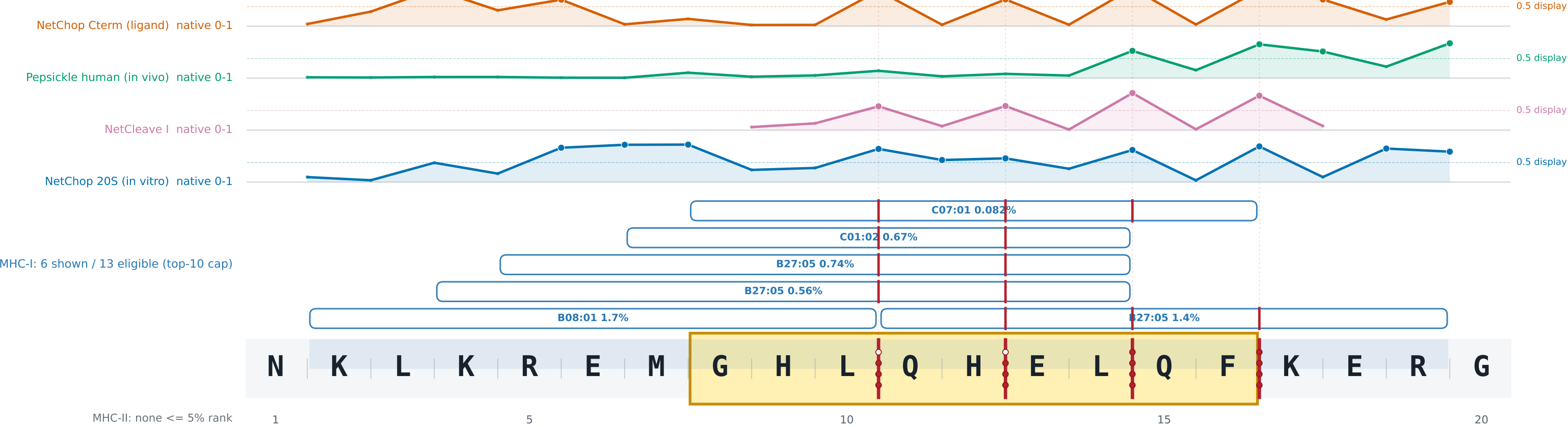
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

WWC1 p.Val179Gly | JLF V3 | 20 aa

disclosed minimal epitope GHLQHELQF at 8-16

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

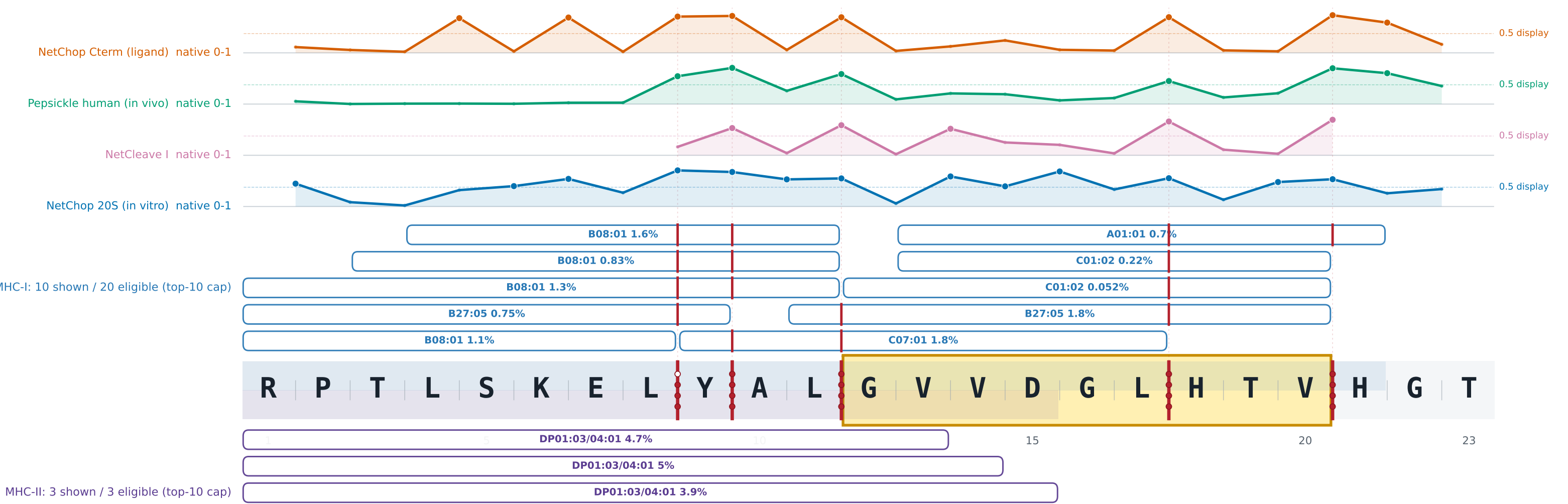
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

BTD p.Phe445Val | JLF V3, JLF V2, JLF V1 | 23 aa

disclosed minimal epitope GVVDGLHTV at 12-20

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

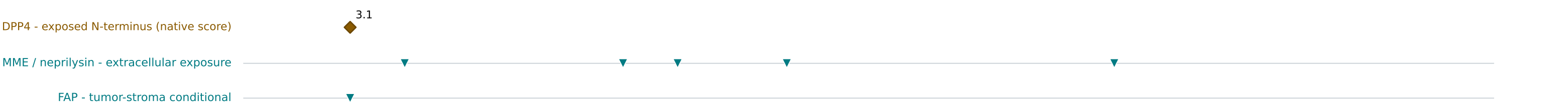


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 1/3, bonds 1-27

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

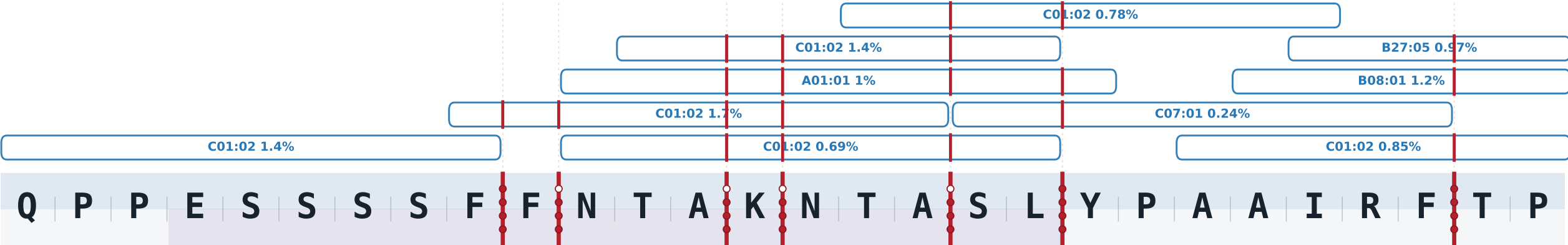
0.5 display

0.5 display

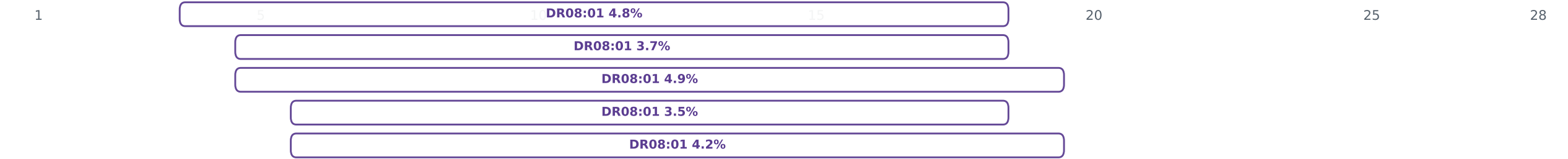
0.5 display

0.5 display

MHC-I: 10 shown / 18 eligible (top-10 cap)



MHC-II: 5 shown / 5 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence

DPP4 - exposed N-terminus (native score)

0.11

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

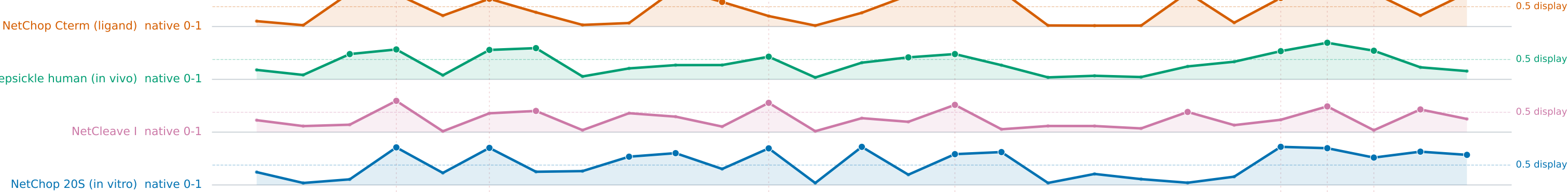
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 2/3, bonds 28-54

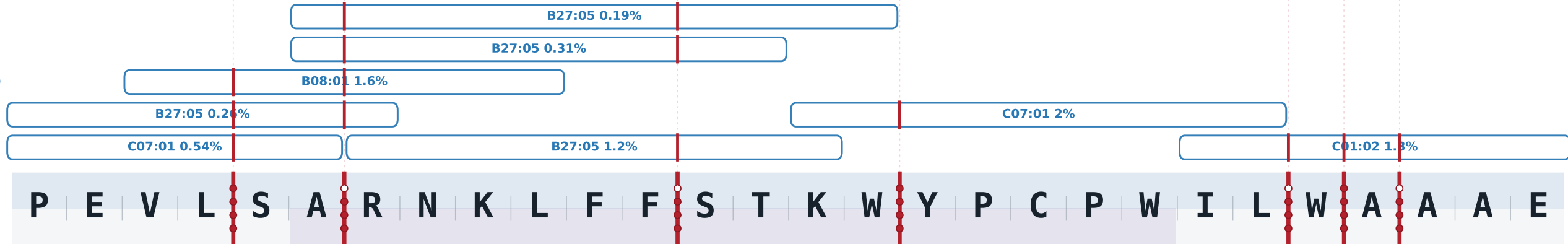
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

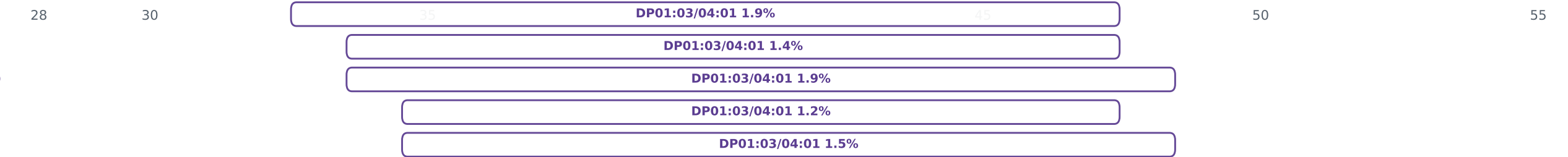
four model-native tracks



MHC-I: 8 shown / 14 eligible (top-10 cap)

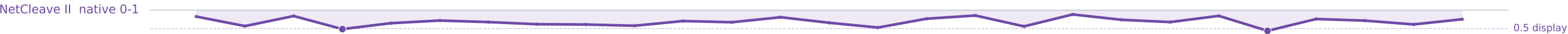


MHC-II: 5 shown / 19 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence

DPP4 - exposed N-terminus (native score)

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

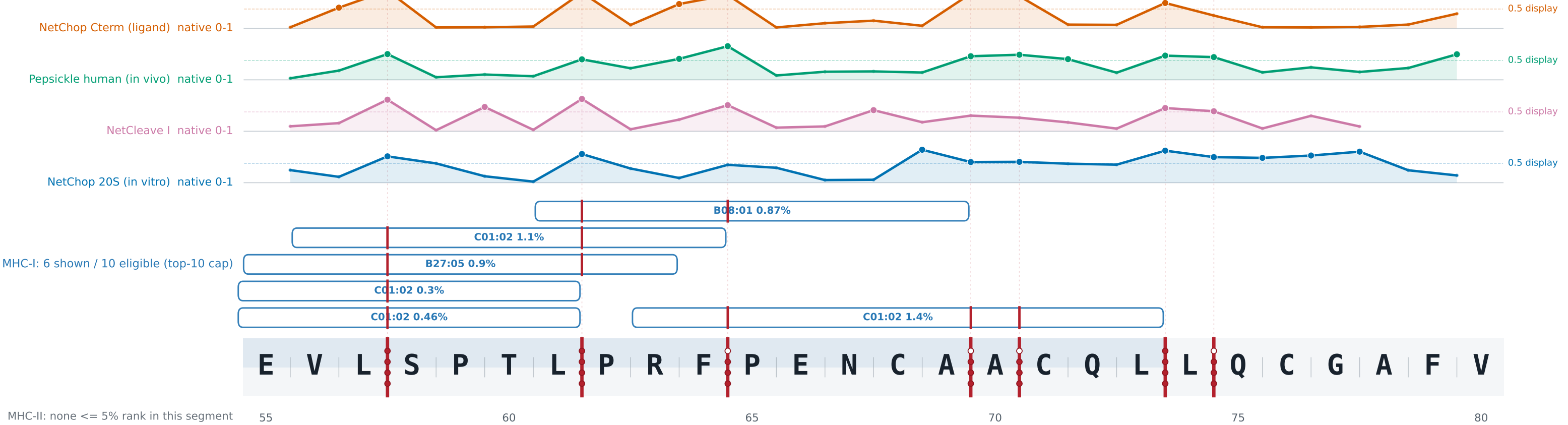
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 3/3, bonds 55-79

minimal epitope/window not disclosed for this SLP

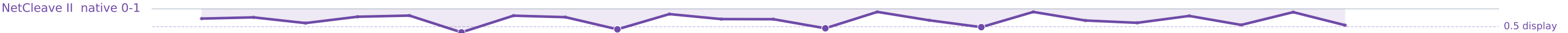
CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence

DPP4 - exposed N-terminus (native score)

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

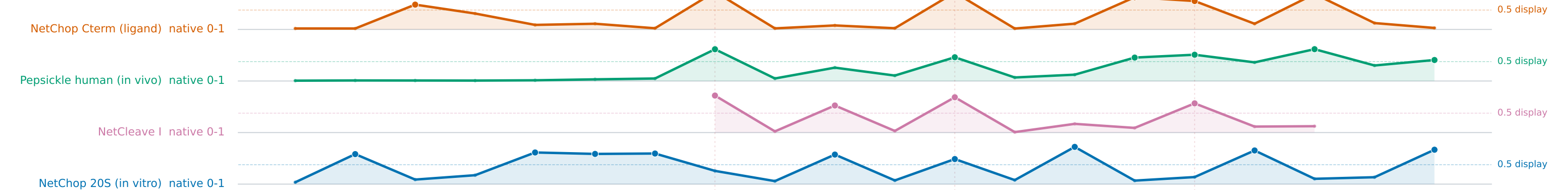
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

KIF1C p.Glu456Lys |JLF V2, JLF V1 | 21 aa

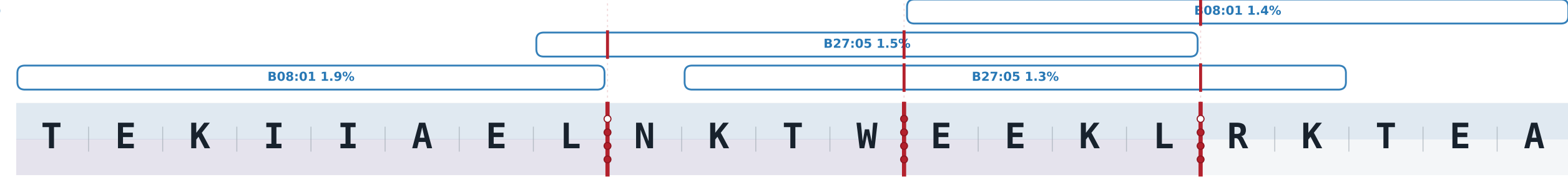
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

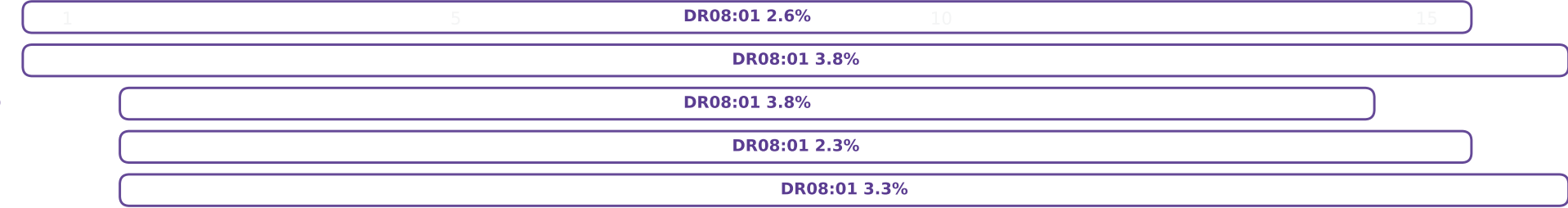
four model-native tracks



MHC-I: 4 shown / 4 eligible (top-10 cap)



MHC-II: 5 shown / 8 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

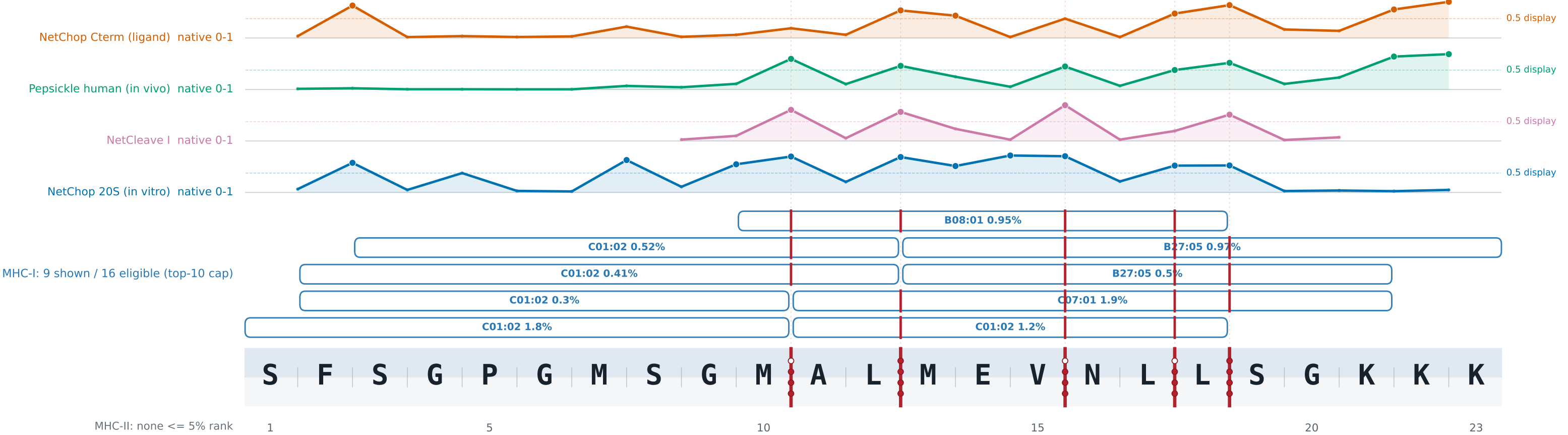
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CD109 p.Arg1310Met | JLF V3 | 23 aa

minimal epitope/window not disclosed for this SLP

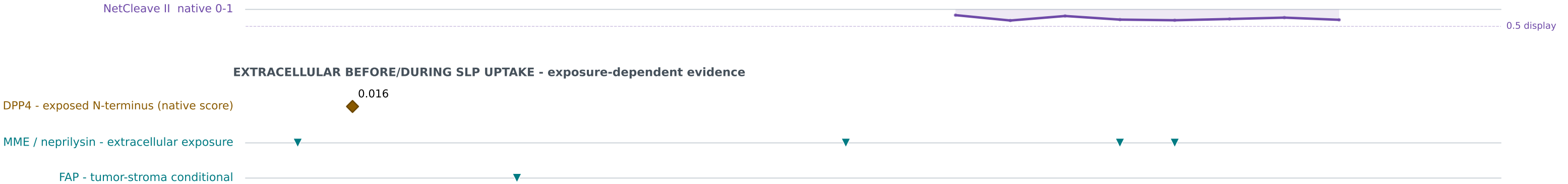
CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

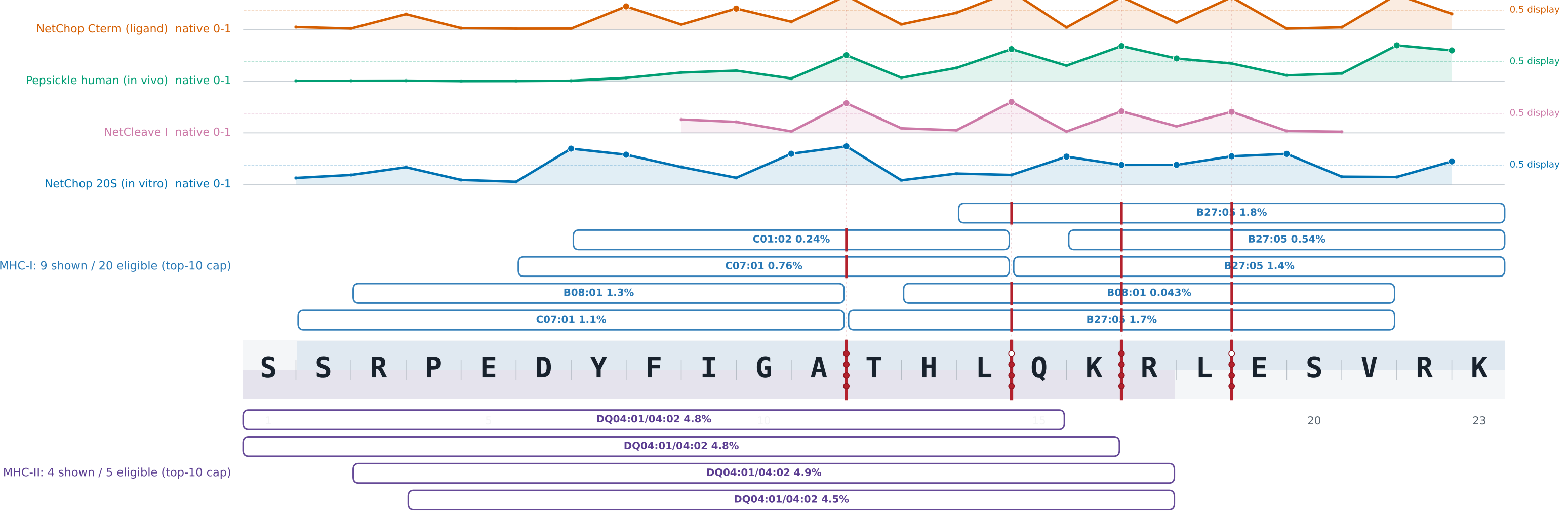
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CENPL p.Pro26His | JLF V3 | 23 aa

minimal epitope/window not disclosed for this SLP

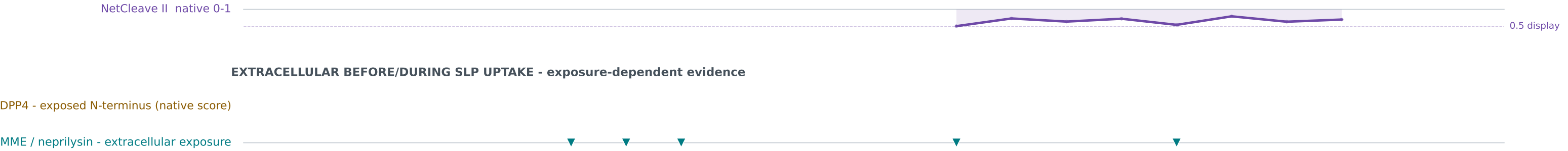
CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

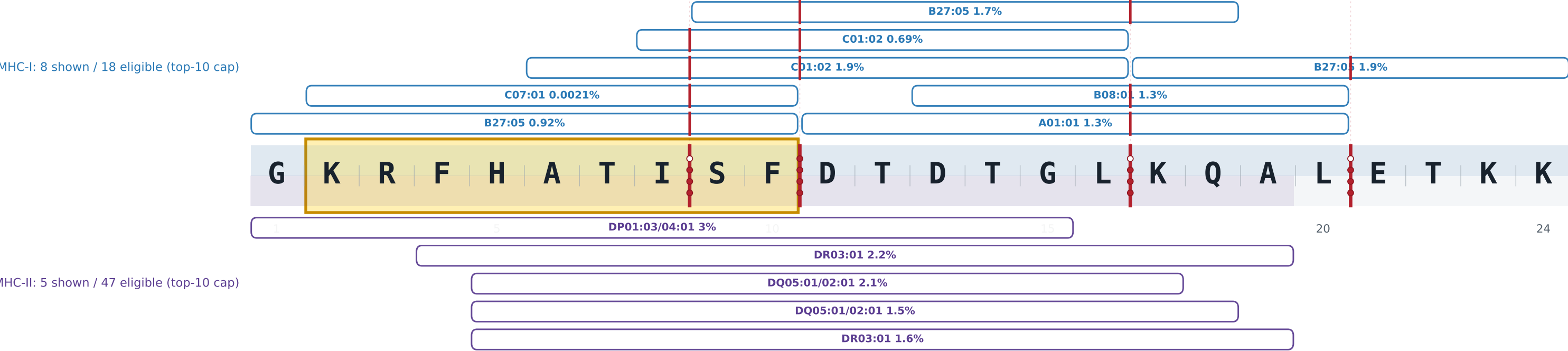
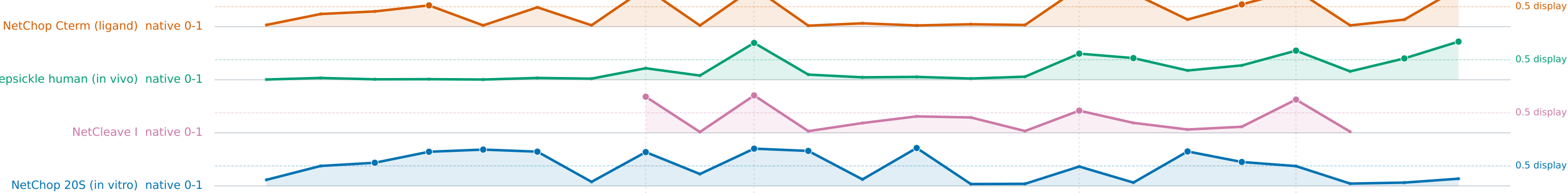
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

DYNC1H1 p.Val314Ile | JLF V3, JLF V2 | 24 aa

disclosed minimal epitope KRFHATISF at 2-10

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence

DPP4 - exposed N-terminus (native score)



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

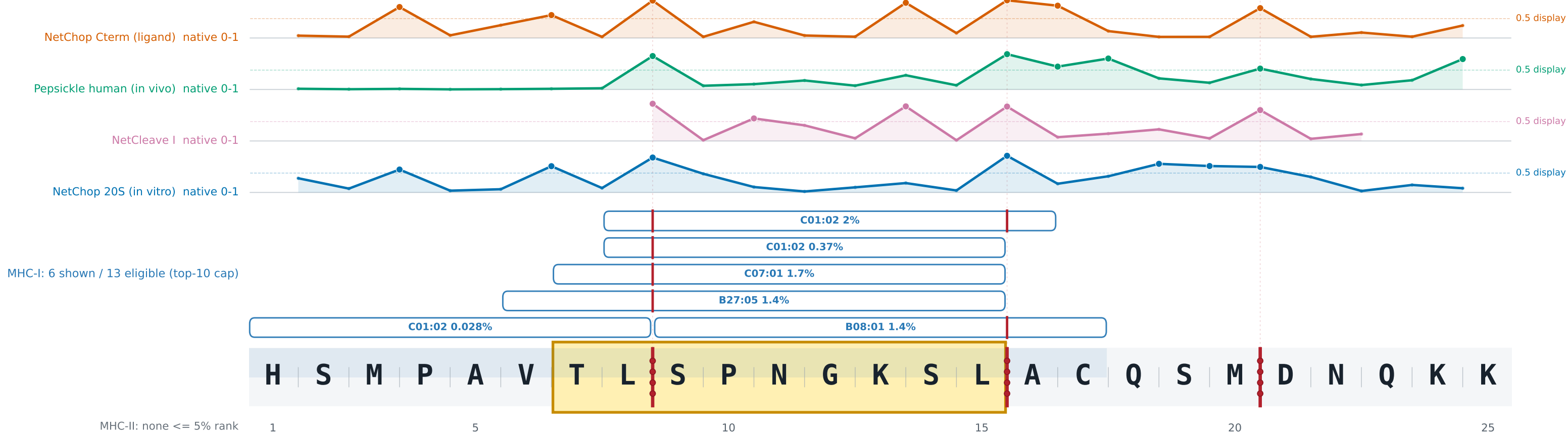
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CDC40 p.Trp475Ser | JLF V3 | 25 aa

disclosed minimal epitope TLSPNGKSL at 7-15

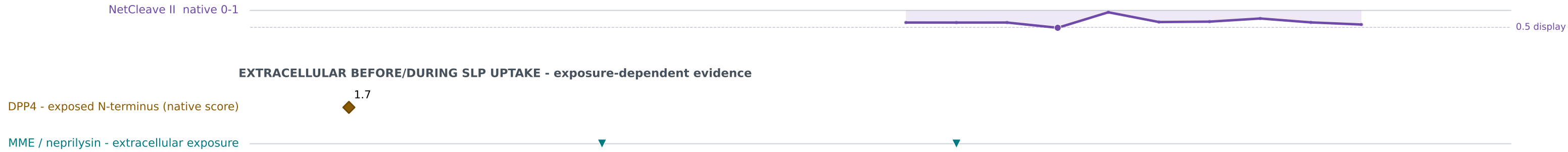
CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

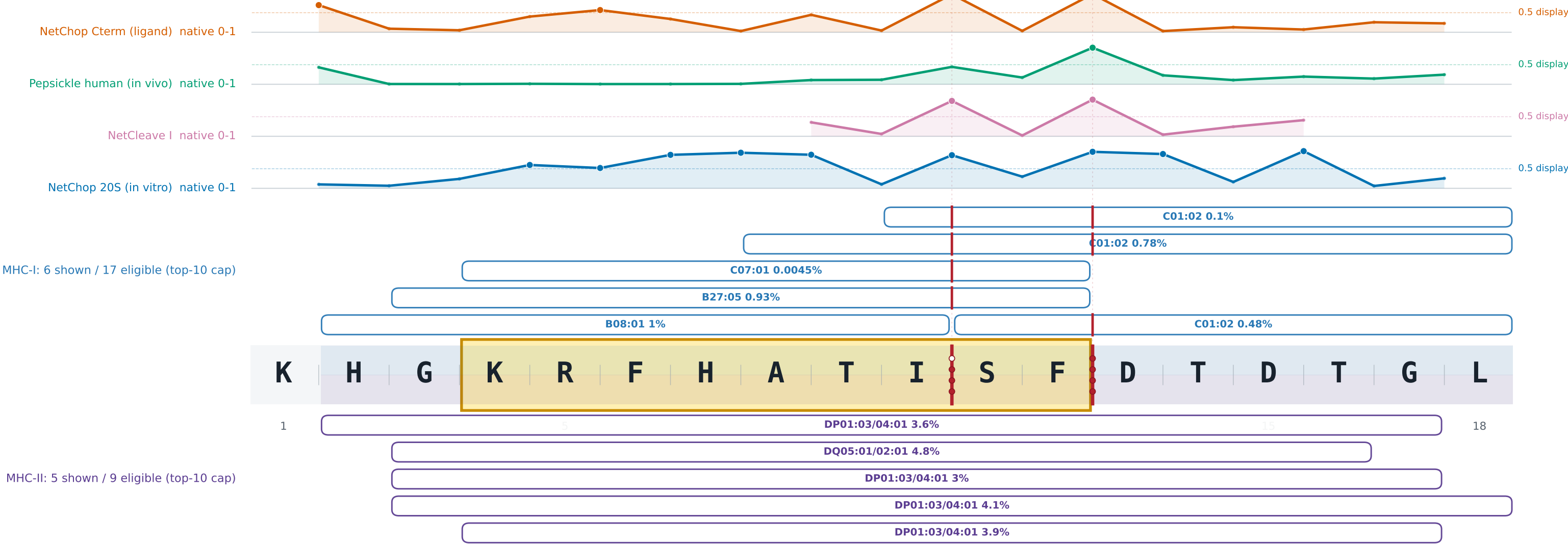
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

DYNC1H1 p.Val314Ile | CeGaT | 18 aa

disclosed minimal epitope KRFHATISF at 4-12

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



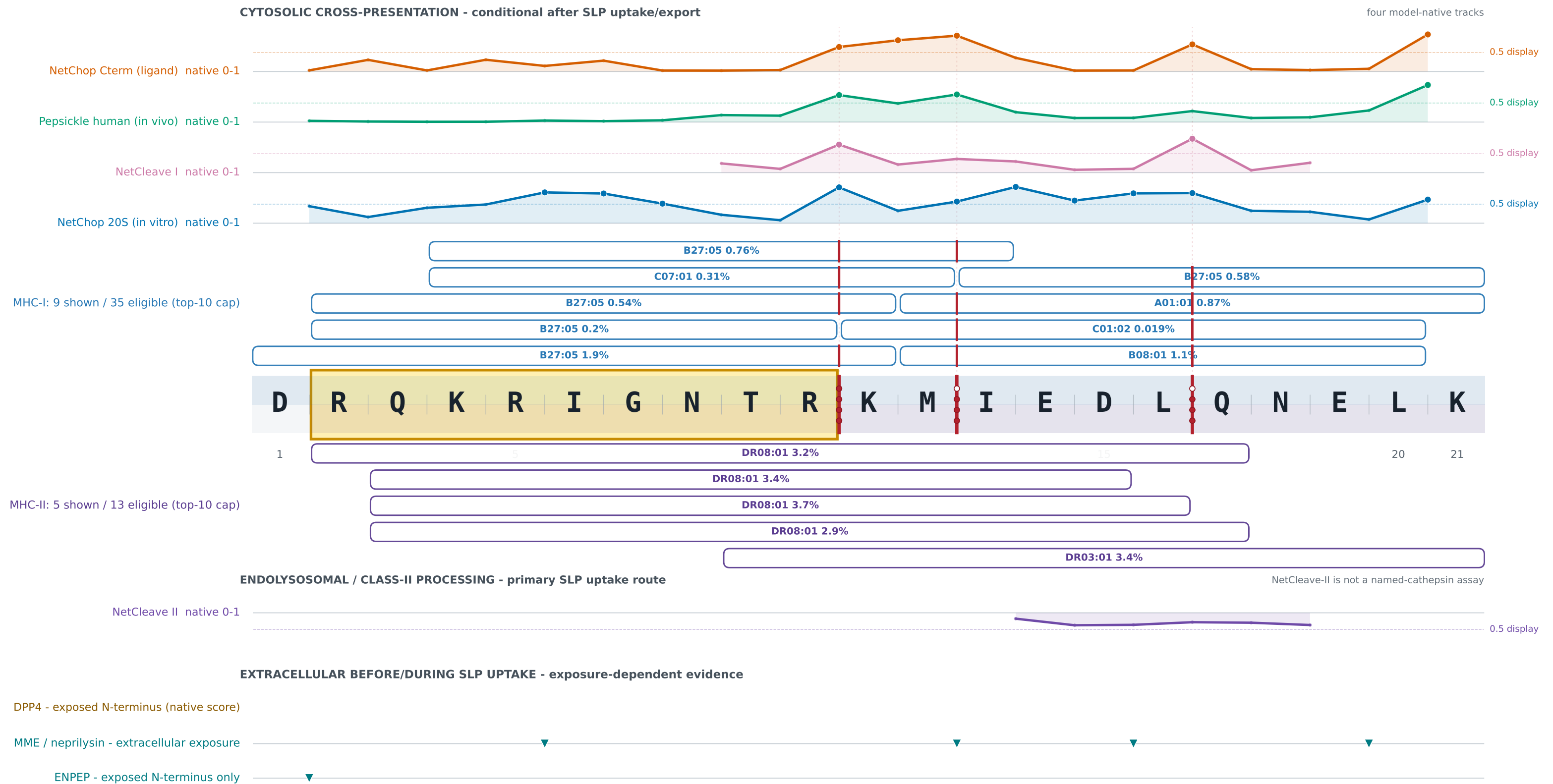
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

SMC5 p.Arg371Lys | JLF V3 | 21 aa

disclosed minimal epitope RQKRIGNTR at 2-10



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

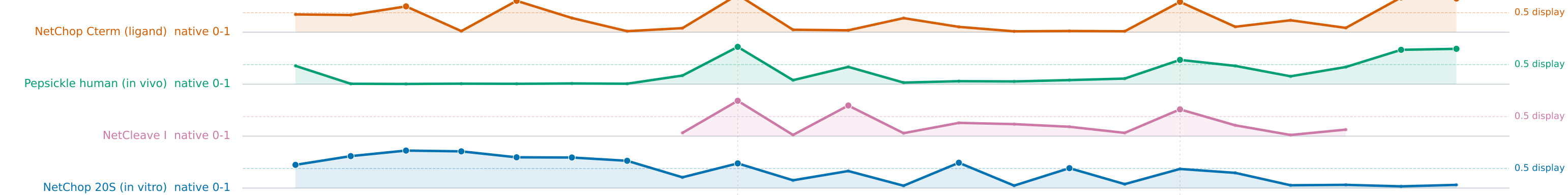
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CUL9 p.Asp403Tyr | JLF V2 | 23 aa

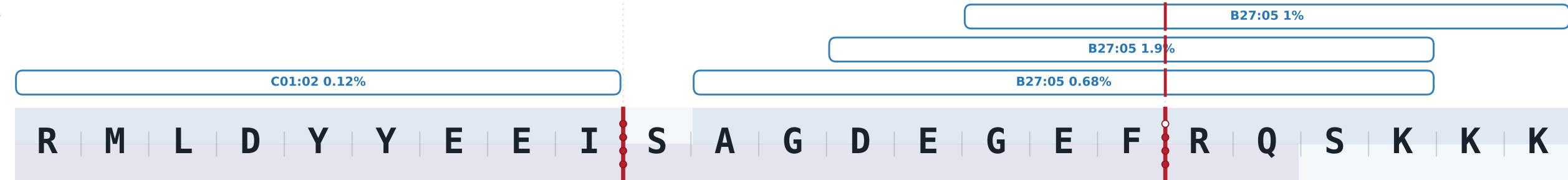
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

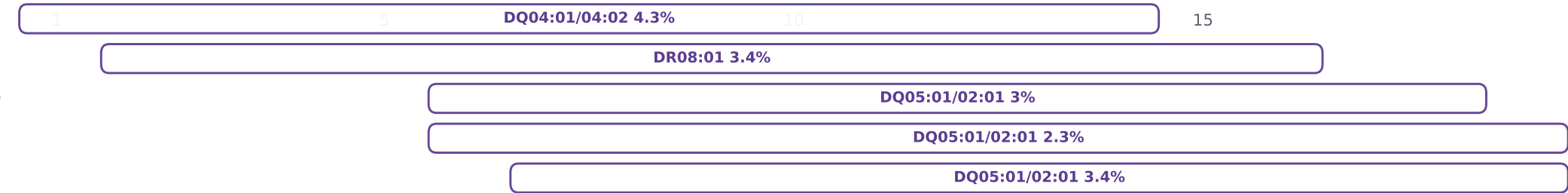
four model-native tracks



MHC-I: 4 shown / 7 eligible (top-10 cap)



MHC-II: 5 shown / 13 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

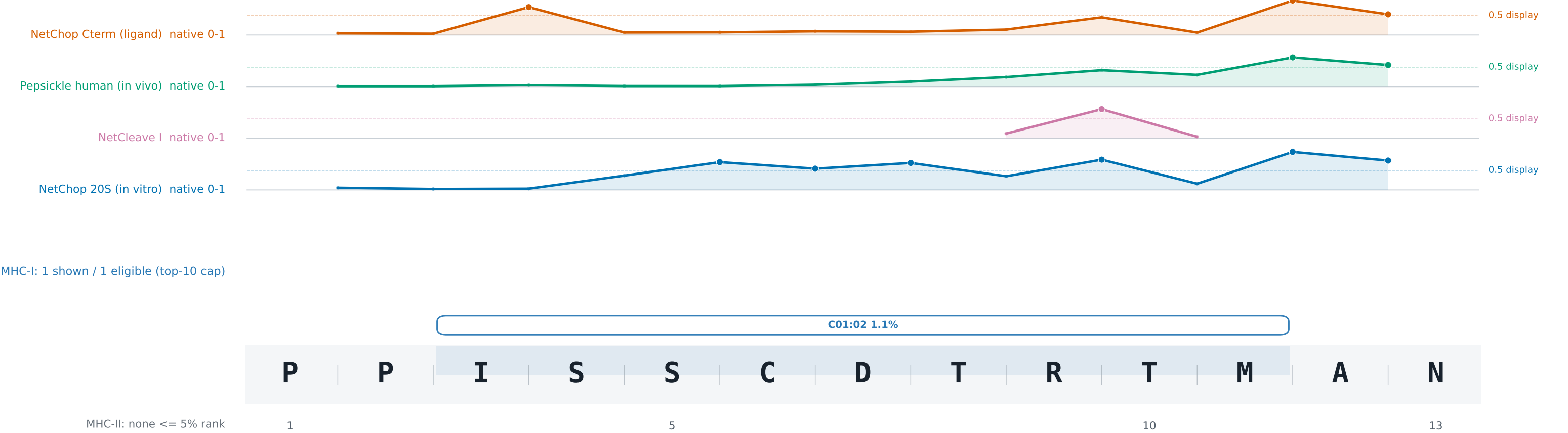
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

NME1 p.Gly24Arg | CeGaT | 13 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence

DPP4 - exposed N-terminus (native score)

MME / neprilysin - extracellular exposure

FAP - tumor-stroma conditional

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

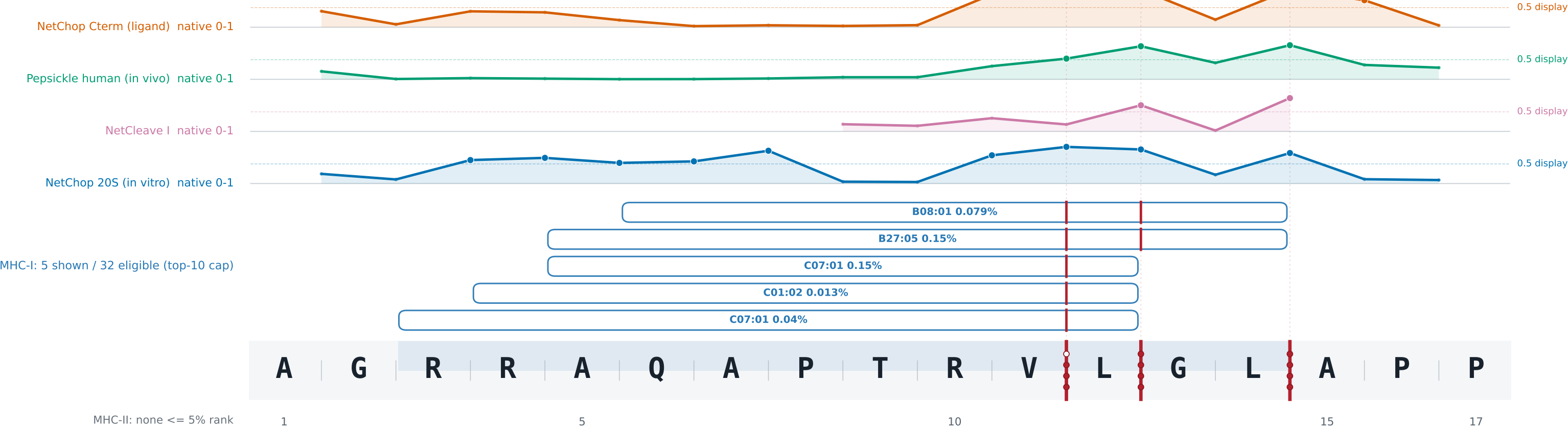
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MYO15B fs | JLF V2, JLF V1 | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

VPS72 p.Lys115Arg | JLF V3 | 27 aa

disclosed minimal epitope ERALLPLEL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

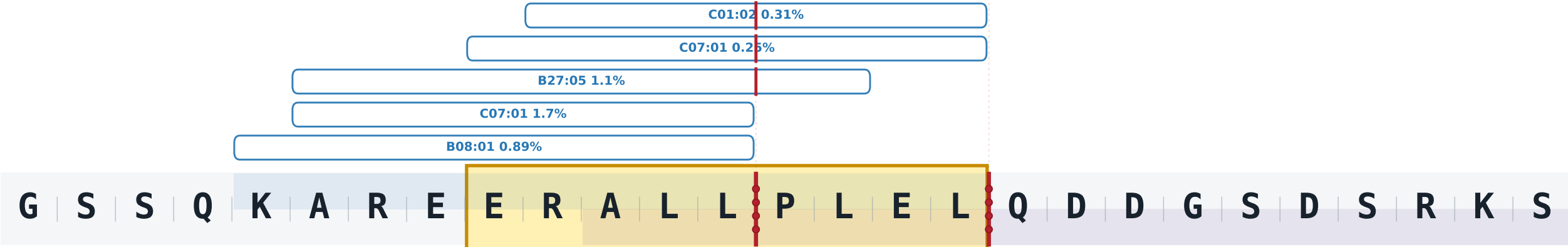
NetCleave I native 0-1

0.5 display

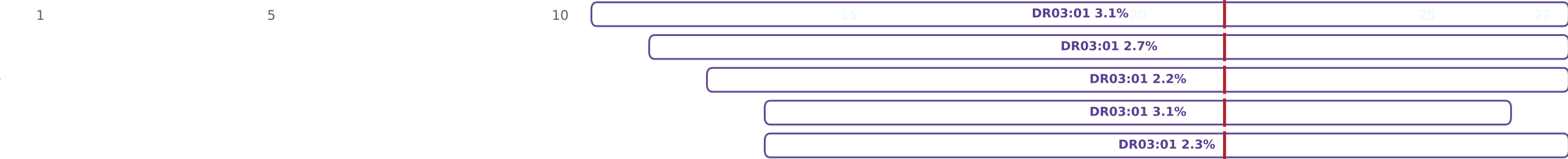
NetChop 20S (in vitro) native 0-1

0.5 display

MHC-I: 5 shown / 12 eligible (top-10 cap)



MHC-II: 5 shown / 10 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence

DPP4 - exposed N-terminus (native score)

0.34

MME / neprilysin - extracellular exposure

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

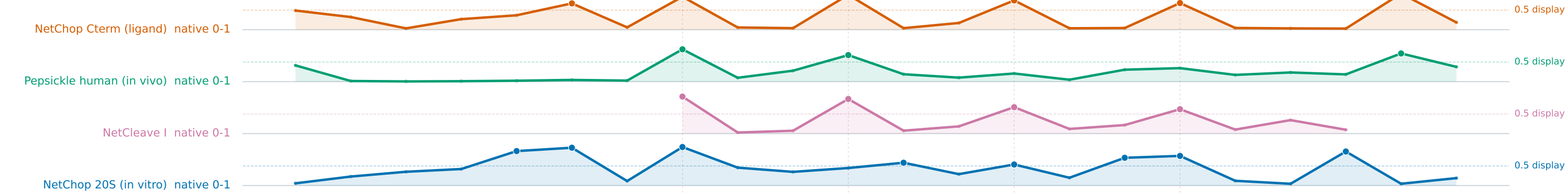
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EXOC4 p.Ser34Ile | JLF V3, JLF V2 | 23 aa

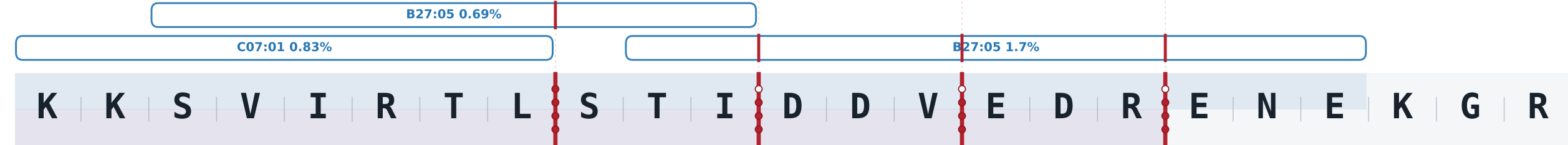
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

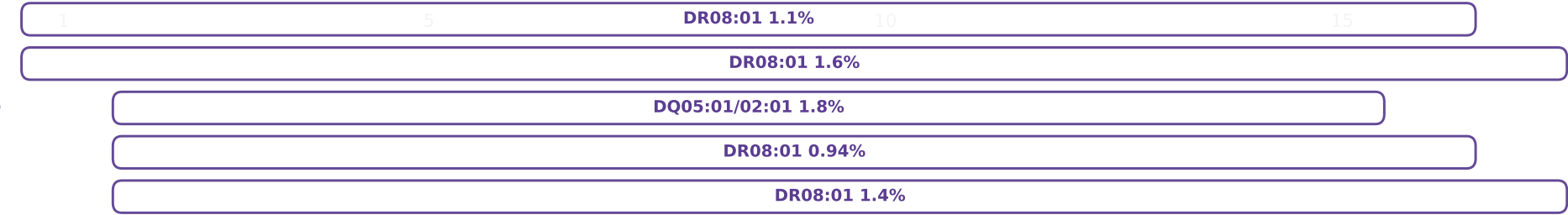
four model-native tracks



MHC-I: 3 shown / 4 eligible (top-10 cap)



MHC-II: 5 shown / 38 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence

DPP4 - exposed N-terminus (native score)



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

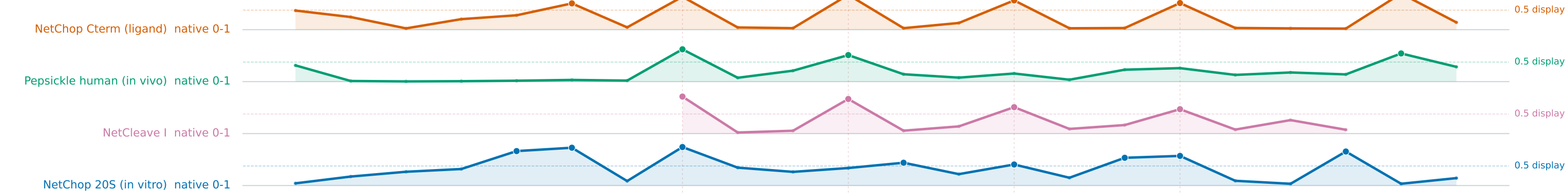
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EPG5 p.Ala869Thr | CeGaT | 23 aa

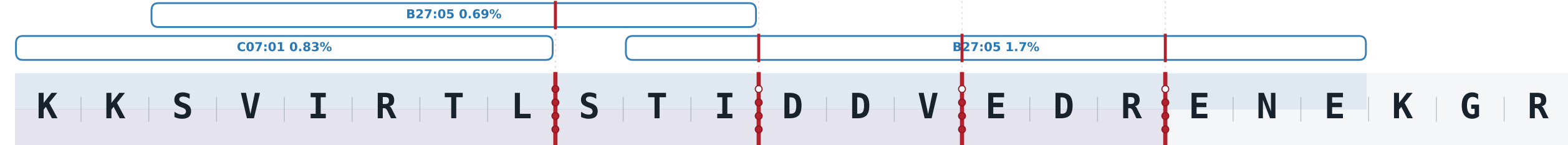
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

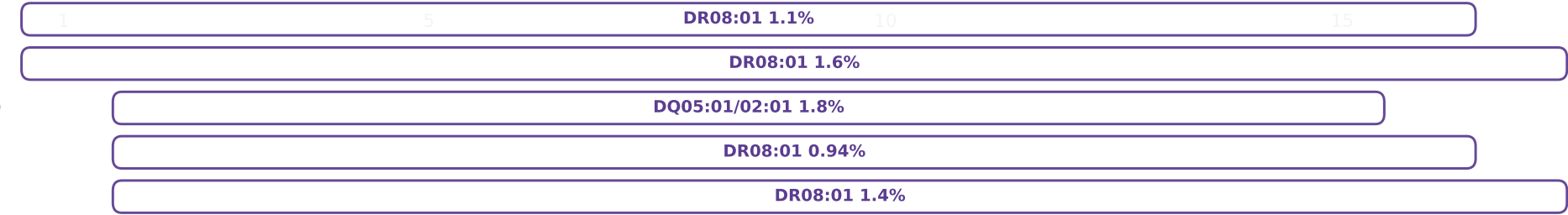
four model-native tracks



MHC-I: 3 shown / 4 eligible (top-10 cap)



MHC-II: 5 shown / 38 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence

DPP4 - exposed N-terminus (native score)



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

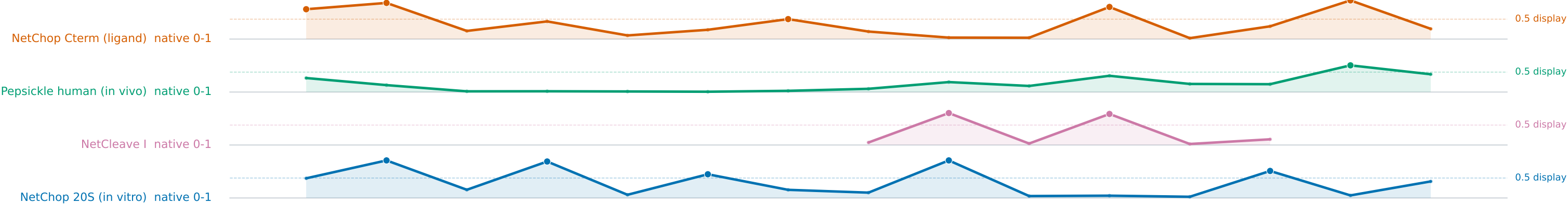
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

TECPR1 p.Thr259fs | CeGaT | 16 aa

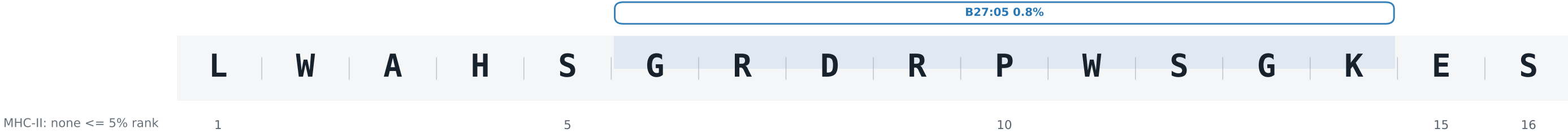
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-I: 1 shown / 1 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

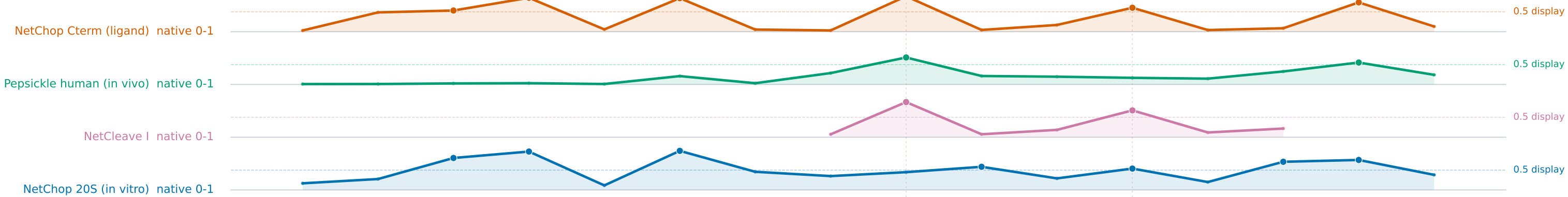
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EXOC4 p.Ser34Ile | CeGaT | 17 aa

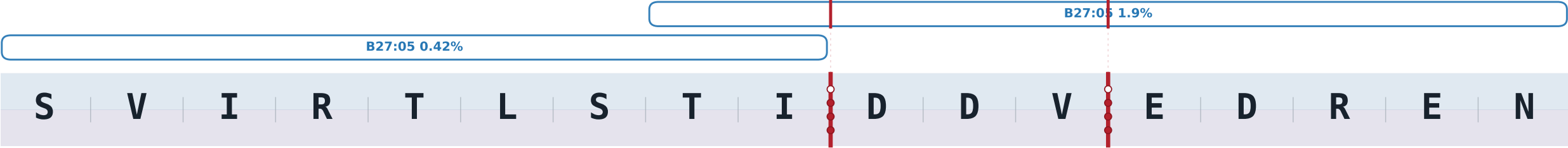
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

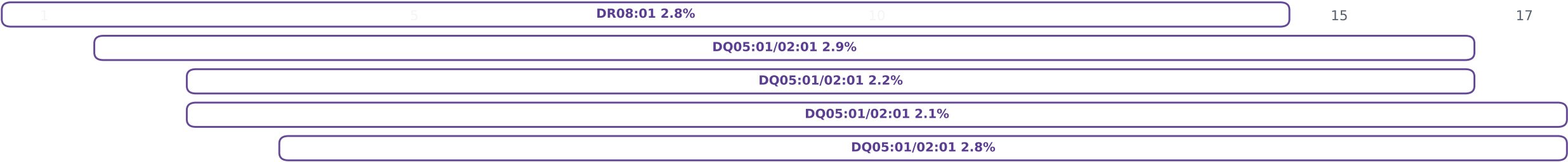
four model-native tracks



MHC-I: 2 shown / 4 eligible (top-10 cap)



MHC-II: 5 shown / 15 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



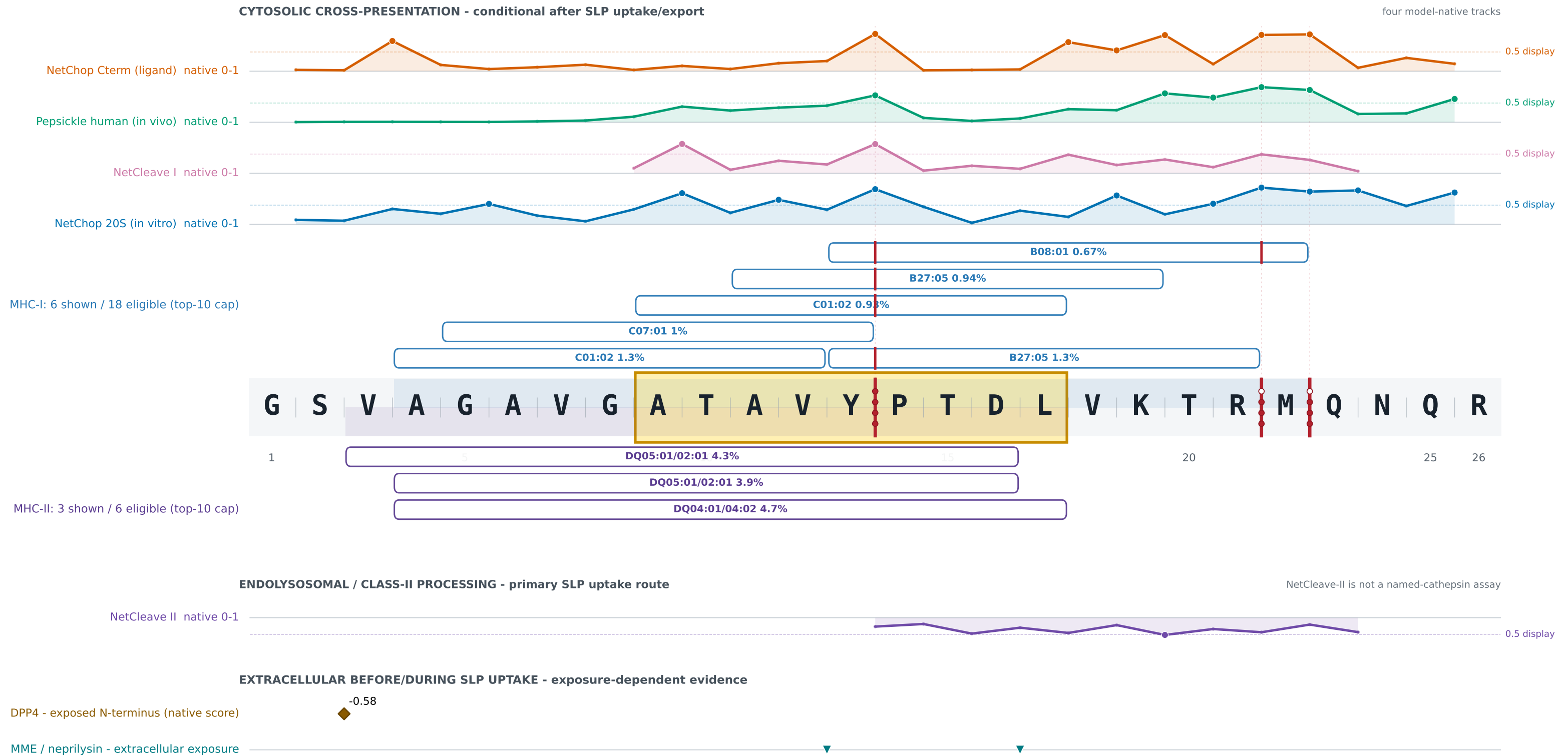
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

SLC25A12 p.Ile347Thr | JLF V2, JLF V1 | 26 aa

disclosed minimal epitope ATAVYPTDL at 9-17



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

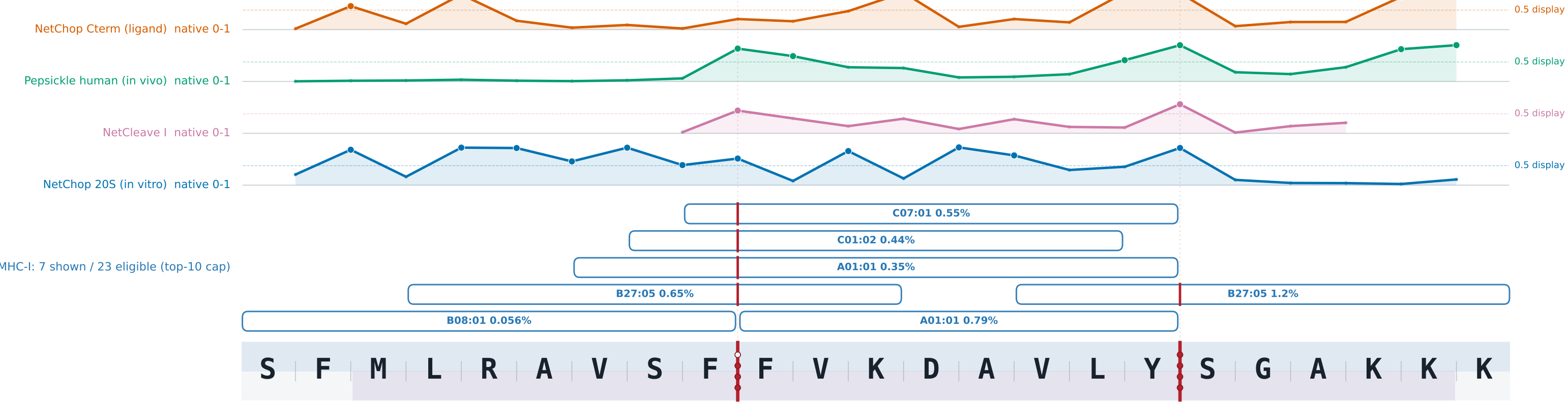
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

PTH1R p.Ile237Phe | JLF V3 | 23 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

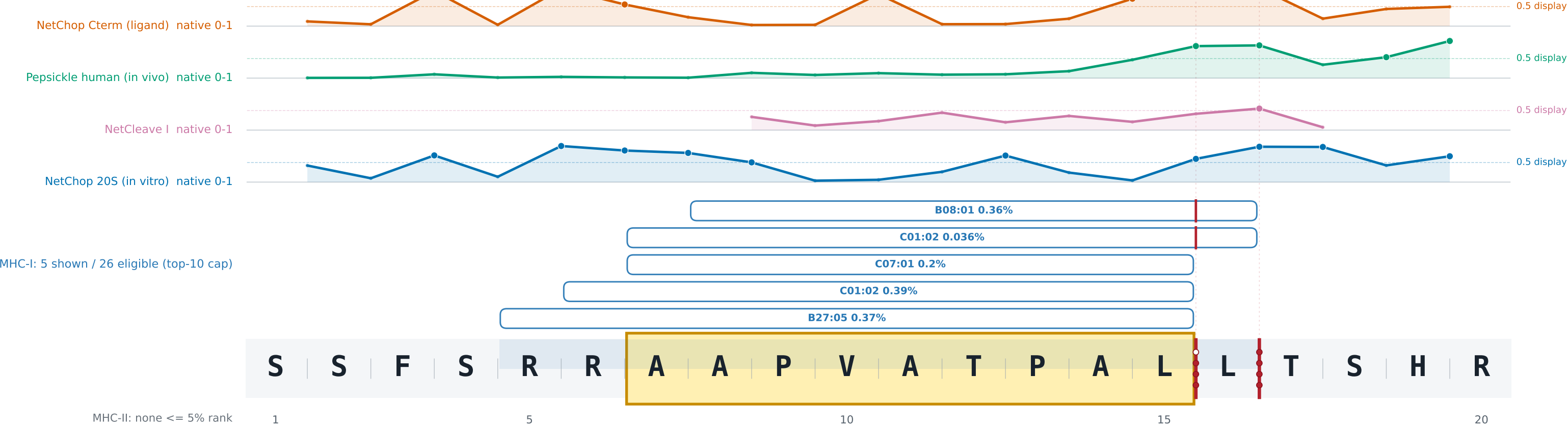
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

PIP5K1A p.Gly474fs | JLF V3 | 20 aa

disclosed minimal epitope AAPVATPAL at 7-15

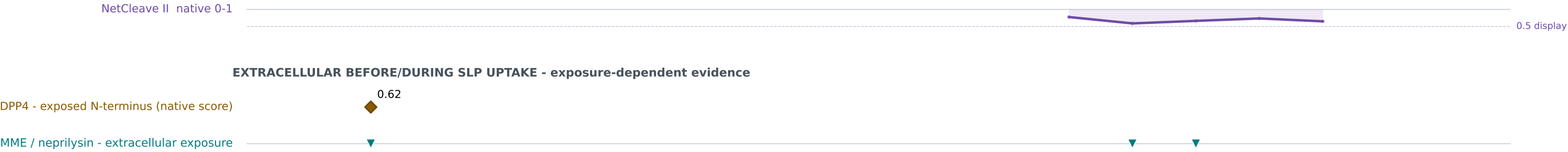
CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

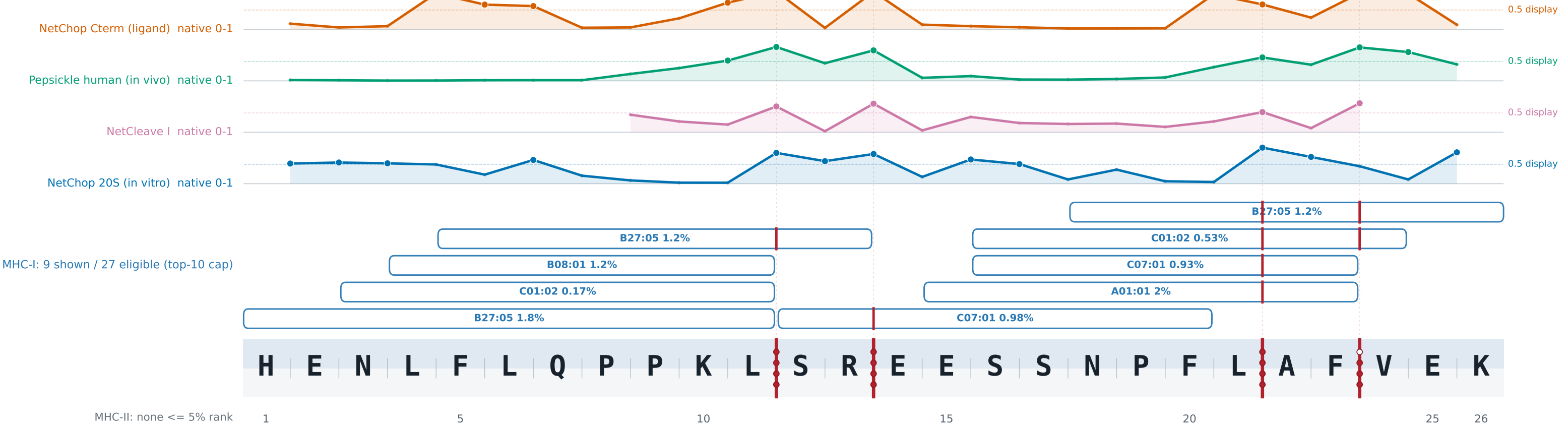
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

KDM3B p.Pro633Ser | JLF V2, JLF V1 | 26 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

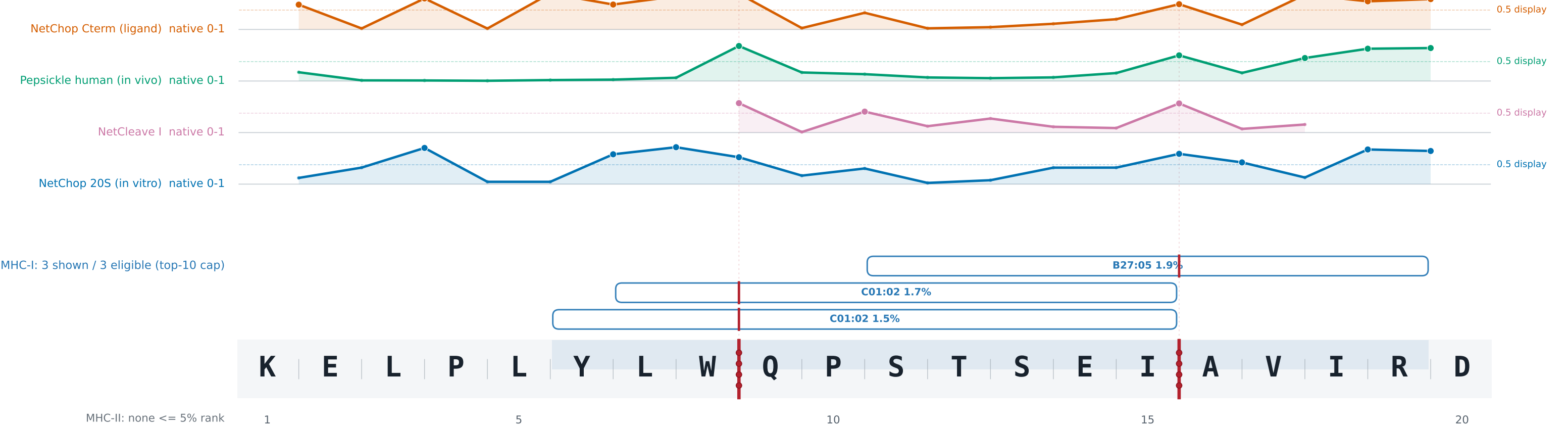
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EPG5 p.Ala869Thr | JLF V1 | 20 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



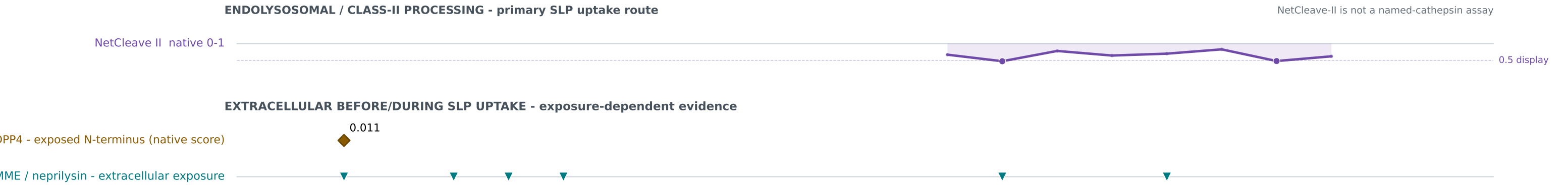
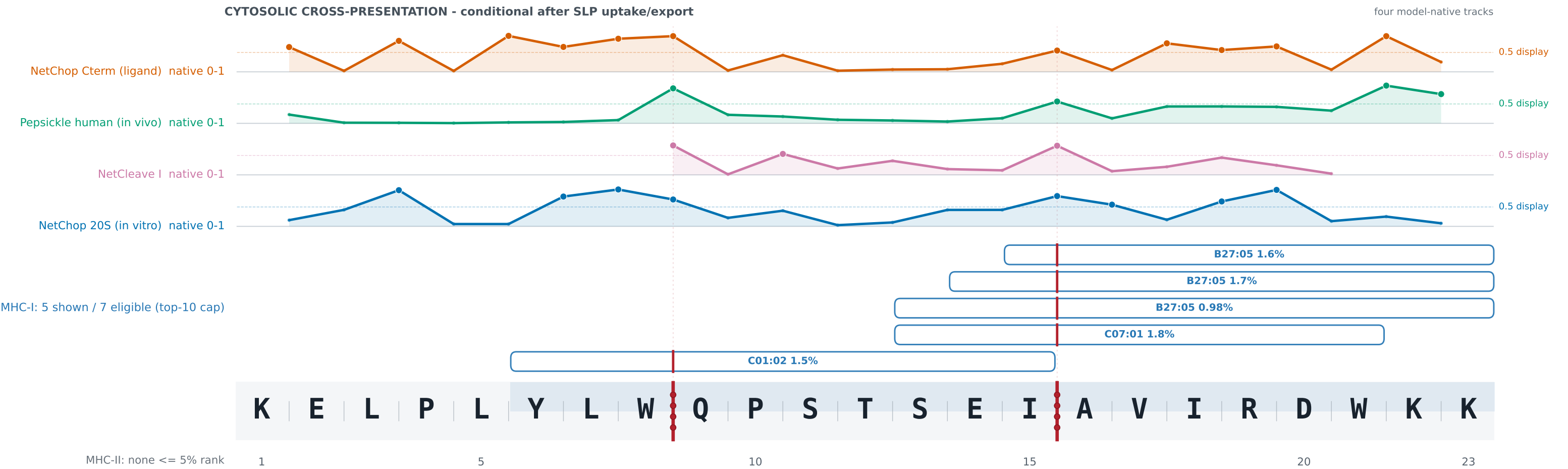
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EPG5 p.Ala869Thr | JLF V2 | 23 aa

minimal epitope/window not disclosed for this SLP



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

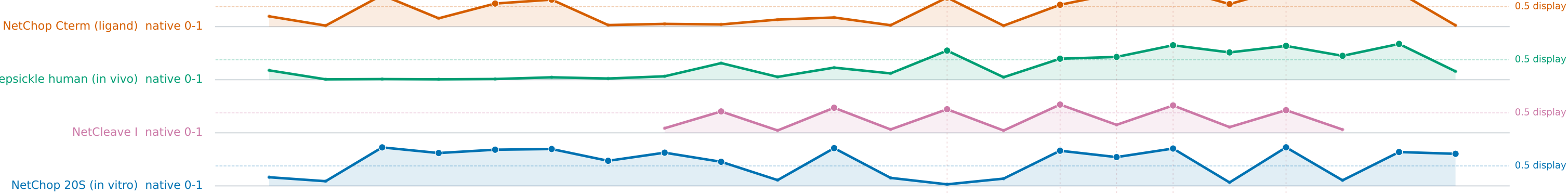
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MAP2 p.Leu867fs | JLF V3 | 23 aa

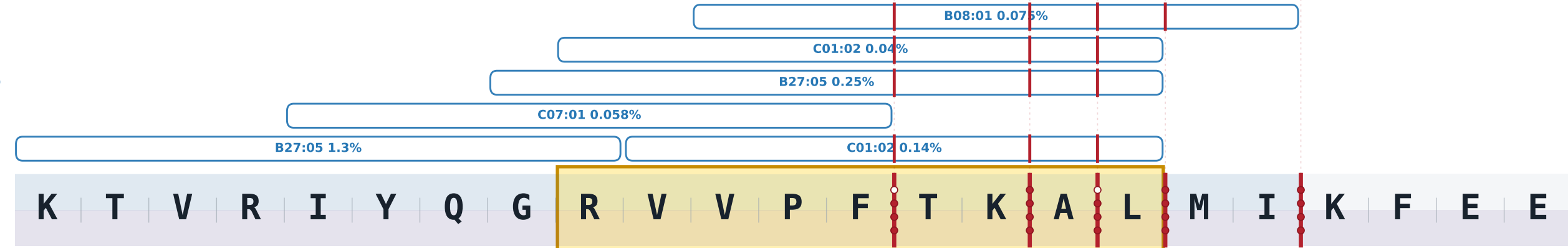
disclosed minimal epitope RVVPFTKAL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-I: 6 shown / 35 eligible (top-10 cap)



MHC-II: 5 shown / 11 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

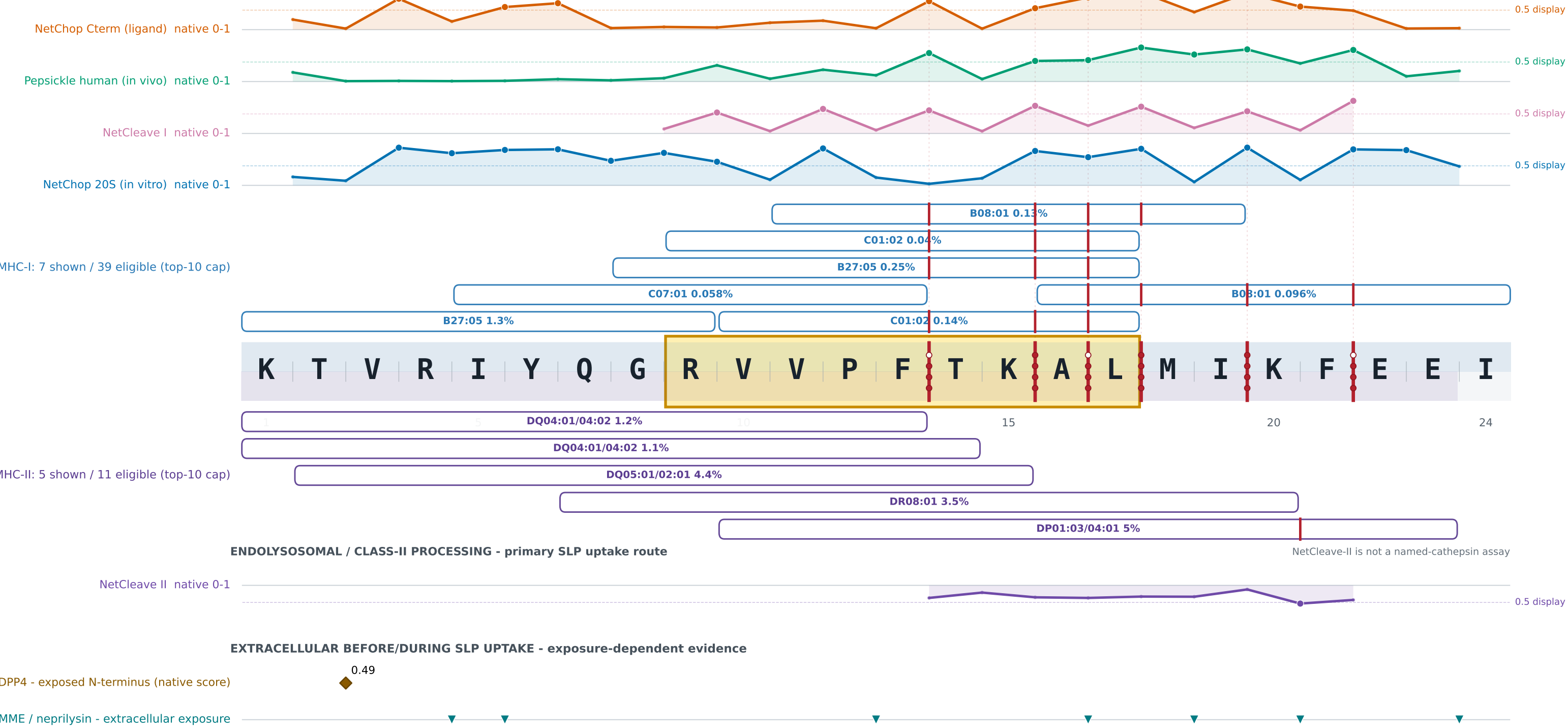
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MAP2 p.Leu867fs | JLF V2, JLF V1 | 24 aa

disclosed minimal epitope RVVPFTKAL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

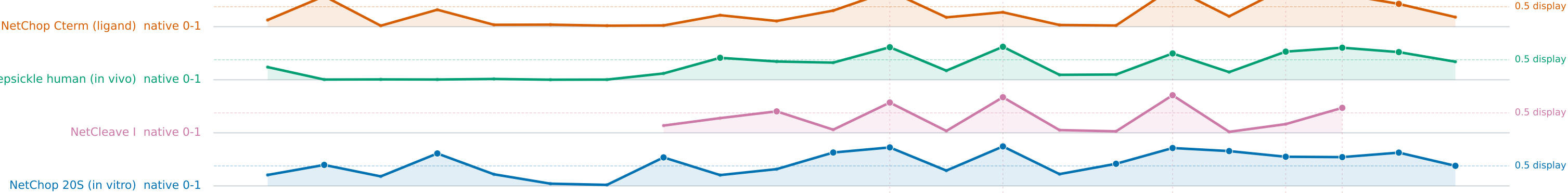
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

BMP1 p.Ser366Thr | JLF V3 | 23 aa

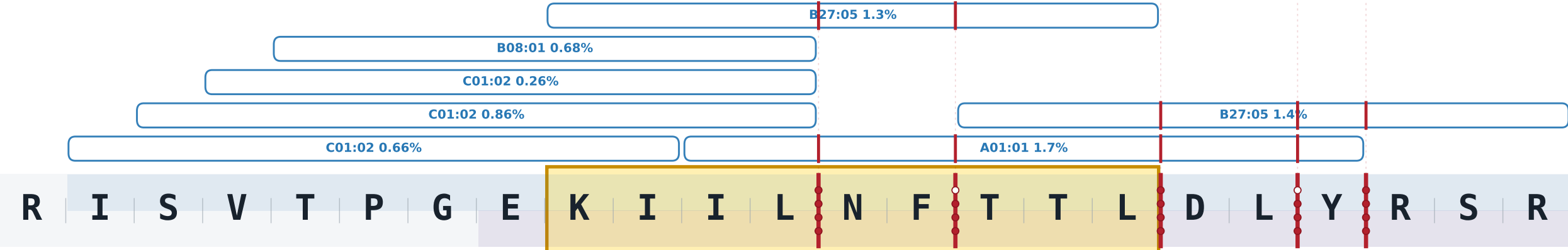
disclosed minimal epitope KIILNFTTL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

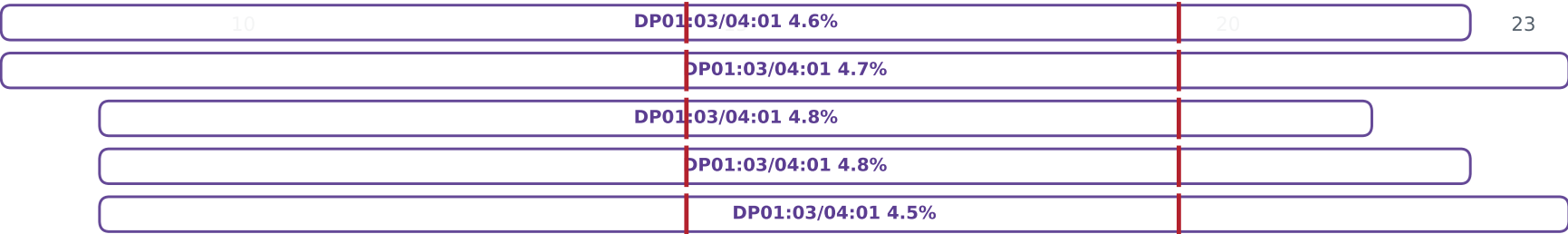
four model-native tracks



MHC-I: 7 shown / 16 eligible (top-10 cap)



MHC-II: 5 shown / 6 eligible (top-10 cap)

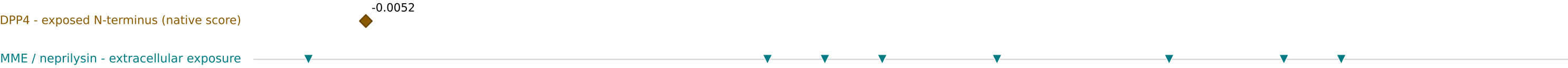


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

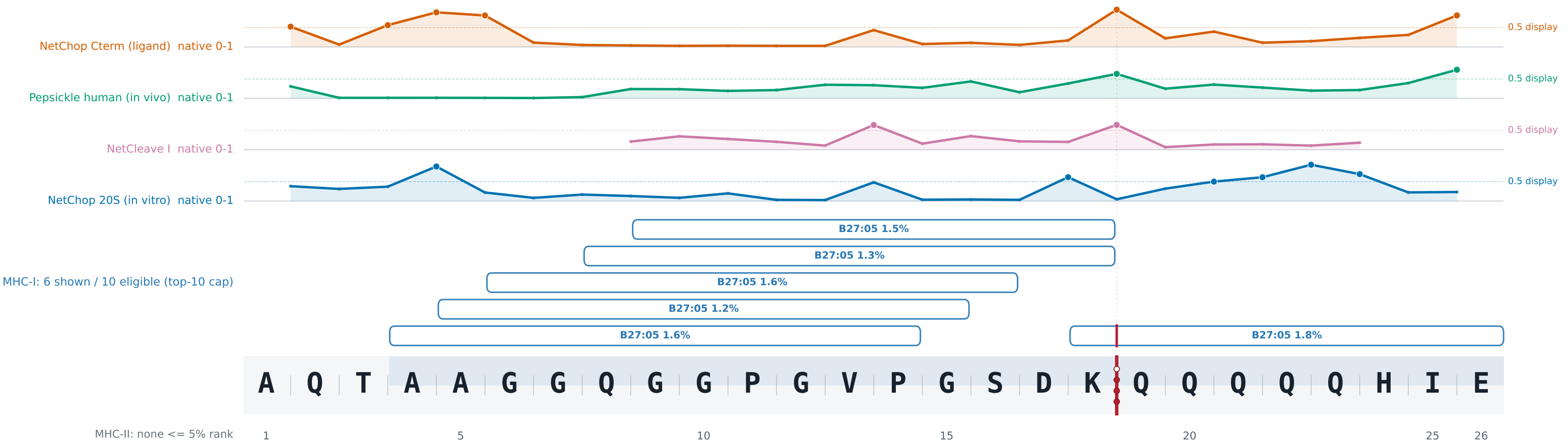
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

NR2F2 p.Gly65Val | JLF V2 | 26 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

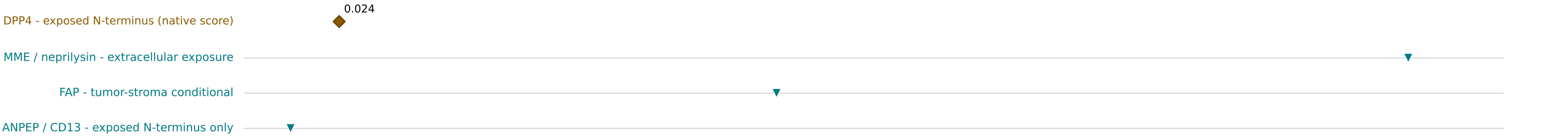


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

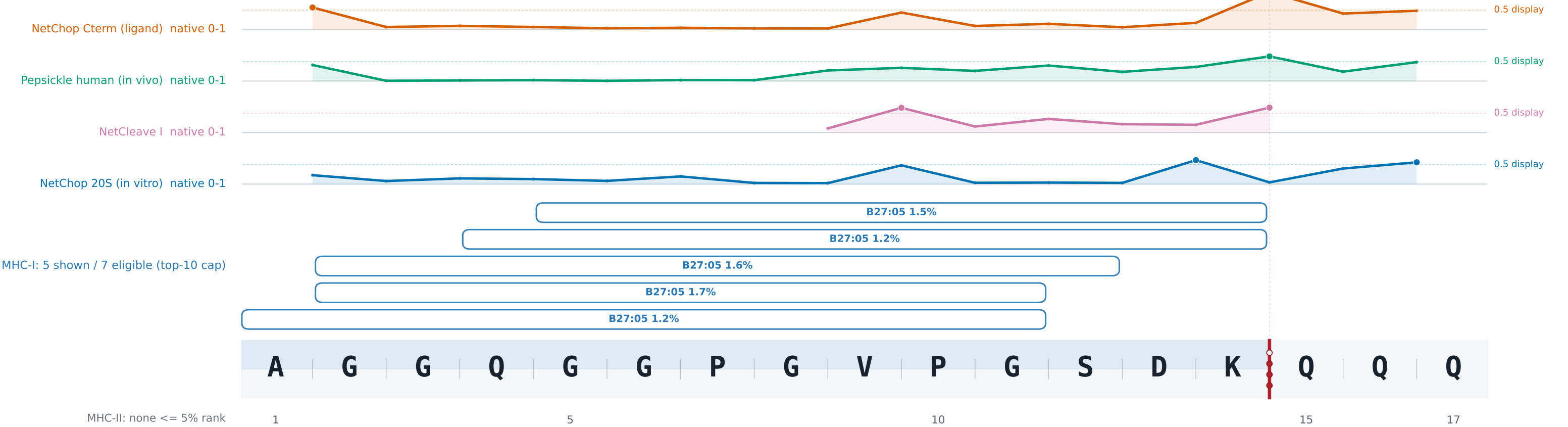
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

NR2F2 p.Gly65Val | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

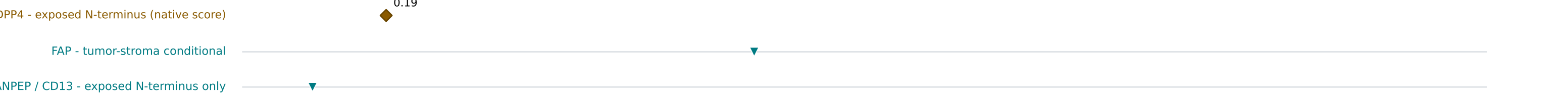


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

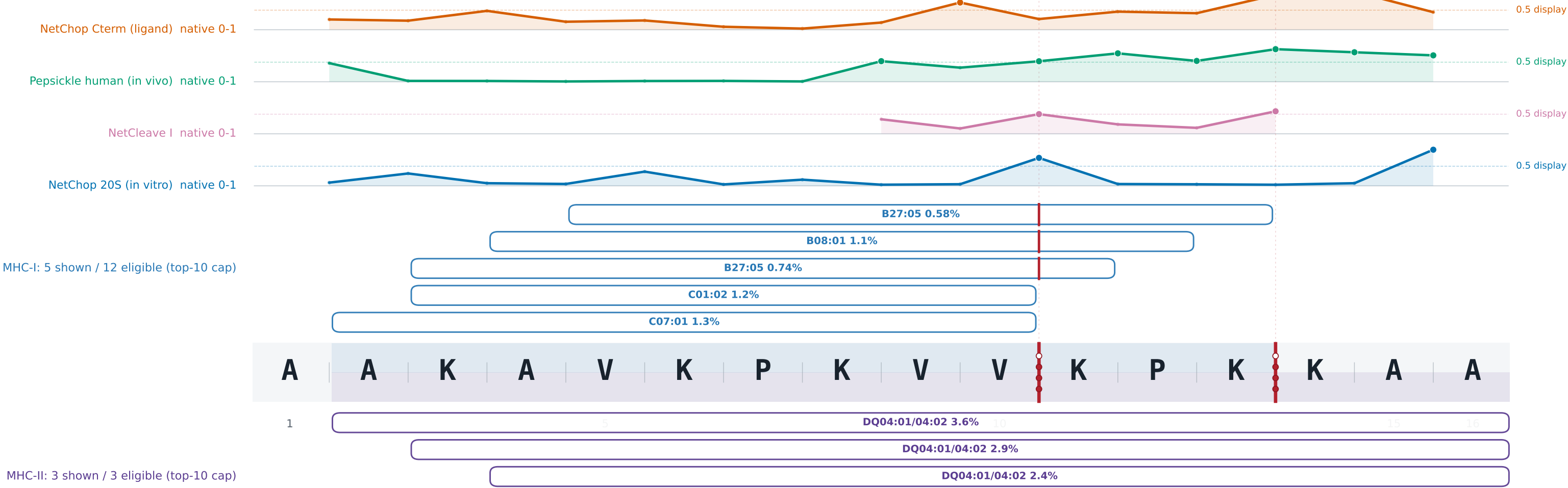
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

H1-2 p.Ala197_Lys201del | CeGaT | 16 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.